

Recursive Macroeconomic Theory

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Complete Solutions

Chapter 8 — Equilibrium with Complete Markets

All 28 end-of-chapter exercises, 8.1–8.28 (pp. 304–330)

What is in this file

The exercises fall into five groups: planners and their decentralisation (8.1–8.3, 8.16, 8.23); pricing in Markov endowment economies with Arrow–Debreu and Arrow securities (8.4–8.8, 8.13, 8.20); corners with linear and logarithmic consumers (8.9, 8.10, 8.27); heterogeneous beliefs (8.14–8.19, 8.21–8.23); and recovering probabilities and discounting from prices (8.11, 8.12, 8.24–8.26, 8.28).

The companion notebook `ls_ch08.ipynb` (same folder) computes every number quoted here and checks each one by an independent route with an `assert`: closed forms against `scipy` optimisation, matrix formulas against brute-force enumeration of histories or truncated sums, and Arrow-security budget constraints state by state. Where the book gives no numbers, the parameters we chose are stated next to their use.

Question boxes reproduce the full statement. Blue boxes are odd-numbered exercises, purple boxes even-numbered ones.

Contents

1 Chapter 8 Equilibrium with Complete Markets	2
1.1 Planners and Decentralisation (8.1–8.3)	3
1.2 Pricing in Endowment Economies (8.4–8.8)	7
1.3 Linear and Logarithmic Consumers: Corners (8.9–8.10)	21
1.4 Changes of Measure (8.11–8.12)	26
1.5 Sequential Trading and Market Structures (8.13–8.14)	27
1.6 Heterogeneous Beliefs (8.15–8.19)	34
1.7 Risk-Free Bonds, Entropy and Beliefs Again (8.20–8.23)	44
1.8 Recovering Probabilities from Prices, and Corners Again (8.24–8.28)	52
Companion code	62

Chapter 8 Equilibrium with Complete Markets

Chapter 8 Overview

Pareto problem (8.4). Maximise $\sum_i \lambda_i U_i(c^i)$, $U_i = \sum_t \sum_{s^t} \beta^t u_i(c_t^i(s^t)) \pi_t(s^t)$, subject to $\sum_i c_t^i(s^t) \leq \sum_i y_t^i(s^t)$. FOC (8.4.2): $\beta^t u_i'(c_t^i(s^t)) \pi_t(s^t) = \lambda_i^{-1} \theta_t(s^t)$. So $u_i'(c_t^i)/u_1'(c_t^1) = \lambda_1/\lambda_i$ (8.4.3): an efficient allocation depends only on the aggregate endowment (Proposition 1).

Time-0 trading (8.5). Budget $\sum_t \sum_{s^t} q_t^0(s^t) c_t^i(s^t) \leq \sum_t \sum_{s^t} q_t^0(s^t) y_t^i(s^t)$ (8.5.1); FOC $\beta^t u_i'(c_t^i(s^t)) \pi_t(s^t) = \mu_i q_t^0(s^t)$ (8.5.4). A competitive equilibrium is the Pareto allocation with $\lambda_i = \mu_i^{-1}$, and $q_t^0 = \theta_t$ (8.5.2). With CRRA utility, $c_t^i = c_t^j (\mu_i/\mu_j)^{-1/\gamma}$ (8.6.1): constant shares. Normalising $q_0^0(s_0) = 1$,

$$q_t^0(s^t) = \beta^t \pi_t(s^t) \frac{u'(c_t^i(s^t))}{u'(c_0^i(s_0))} \quad \text{for any } i. \quad (*)$$

Asset pricing (8.7). A claim to $\{d_t(s^t)\}$ costs $\sum_t \sum_{s^t} q_t^0(s^t) d_t(s^t)$ (8.7.1). Prices in units of the date- τ good: $q_t^\tau(s^t) = q_t^0(s^t)/q_\tau^0(s^\tau)$ (8.7.5).

Sequential trading (8.8). Budget $\tilde{c}_t^i + \sum_{s_{t+1}} \tilde{a}_{t+1}^i(s_{t+1}, s^t) \tilde{Q}_t(s_{t+1}|s^t) \leq y_t^i(s^t) + \tilde{a}_t^i(s^t)$ (8.8.3); natural debt limit $A_t^i(s^t) = \sum_{\tau \geq t} \sum_{s^\tau|s^t} q_\tau^t(s^\tau) y_\tau^i(s^\tau)$ (8.8.2), $-\tilde{a}_{t+1}^i \leq A_{t+1}^i$ (8.8.4). Kernel $\tilde{Q}_t(s_{t+1}|s^t) = \beta \pi_t(s^{t+1}|s^t) u_i'(c_{t+1}^i(s_{t+1}|s^t))/u_i'(c_t^i(s^t))$ (8.8.6). The time-0 allocation is a sequential equilibrium with zero initial wealth and holdings $\tilde{a}_t^i(s^t) = \Upsilon_t^i(s^t)$, $\Upsilon_t^i(s^t) = \sum_{\tau \geq t} \sum_{s^\tau|s^t} q_\tau^t(s^\tau) [c_\tau^i(s^\tau) - y_\tau^i(s^\tau)]$ (8.8.1).

Markov case (8.9–8.10). With $y_t^i = y^i(s_t)$: $c_t^i = \bar{c}^i(s_t)$, $Q(s'|s) = \beta u_i'(\bar{c}^i(s')) \pi(s'|s)/u_i'(\bar{c}^i(s))$ (8.9.4), debt limits $\bar{A}^i(s) = y^i(s) + \sum_{s'} Q(s'|s) \bar{A}^i(s')$ (8.9.13), and j -step kernels $Q_j(s_{t+j}|s_t) = \sum_{s_{t+1}} Q_1(s_{t+1}|s_t) Q_{j-1}(s_{t+j}|s_{t+1})$ (8.10.3). In matrix form, with $Q_{ij} = Q(s_{t+1}=j|s_t=i)$, $Q_j = Q^j$ and $\bar{A}^i = (I - Q)^{-1} y^i$.

Heterogeneous beliefs (appendix B). With subjective π_t^i , the Pareto FOC becomes $\frac{\pi_t^i(s^t) u_i'(c_t^i)}{\pi_t^1(s^t) u_1'(c_t^1)} = \frac{\lambda_1}{\lambda_i}$ (8.B.2). With i.i.d. truth π , the consumer whose beliefs have the smallest relative entropy $\text{ent}(\pi, \pi^i) = \sum_s \pi(s) \log[\pi(s)/\pi^i(s)]$ eventually consumes everything (8.B.4).

Conventions used throughout

Matrices are indexed by states in the order listed in each exercise. e_i is the i -th unit vector and $\mathbf{1}$ a vector of ones. For a Markov chain, P^t gives t -step transition probabilities, so $\sum_{s^t} \pi_t(s^t) f(s_t) = e'_{s_0} P^t f$. We use $\sum_t \beta^t P^t = (I - \beta P)^{-1}$ throughout. Prices are in units of the time-0 good ($q_0^0(s_0) = 1$) unless stated.

Planners and Decentralisation (8.1–8.3)

Exercise 8.1 Family economics, I

Question

p. 304. There is one consumption good and one input, labor. A family has two members named 1 and 2. The family is run by person 1. His welfare function is

$$\lambda_1 \log c_1 + \lambda_2 [\log c_2 - n_2],$$

where λ_1 and λ_2 are positive Pareto weights, c_1, c_2 are consumption of person 1 and person 2, respectively, and n_2 is labor supplied by person 2. Person 2 has no labor but is endowed with s units of the consumption good, where $s > 0$. Feasible allocations satisfy

$$c_1 + c_2 \leq n_2 + s.$$

- a. Formulate the Pareto problem as a Lagrangian.
- b. Solve the Pareto problem for an optimal allocation and Lagrange multiplier.
- c. Interpret the Lagrange multiplier as a “shadow price”, and tell the object of which it is the shadow price.
- d. Describe how the family could be reorganized as a competitive economy, being careful to identify an initial distribution of property and a price system.
- e. Compute a competitive equilibrium of the economy that you identified in part d.
- f. Please tell how n_2 would respond to different values of s .

Solution:

Misprint. “Person 2 has no labor” contradicts “ n_2 is labor supplied by person 2”. It should read *person 1* has no labor but is endowed with s . Exercise 8.2 confirms this: “instead of being endowed with s units of consumption, household 1 is endowed with 1 unit”. The Pareto problem does not depend on who owns s ; parts (d)–(f) use this reading.

(a) With multiplier θ on the resource constraint and the restriction $n_2 \geq 0$,

$$\mathcal{L} = \lambda_1 \log c_1 + \lambda_2 (\log c_2 - n_2) + \theta (n_2 + s - c_1 - c_2).$$

(b) First-order conditions:

$$c_1 : \frac{\lambda_1}{c_1} = \theta, \quad c_2 : \frac{\lambda_2}{c_2} = \theta, \quad n_2 : -\lambda_2 + \theta \leq 0 \quad (= 0 \text{ if } n_2 > 0).$$

Interior case ($n_2 > 0$): $\theta = \lambda_2$, so $c_1 = \lambda_1/\lambda_2$ and $c_2 = 1$. The resource constraint binds ($\theta > 0$): $n_2 = c_1 + c_2 - s$. Hence

$$c_1 = \frac{\lambda_1}{\lambda_2}, \quad c_2 = 1, \quad n_2 = 1 + \frac{\lambda_1}{\lambda_2} - s, \quad \theta = \lambda_2,$$

valid when $n_2 > 0$, i.e. $s < 1 + \lambda_1/\lambda_2$. *Corner* ($s \geq 1 + \lambda_1/\lambda_2$): $n_2 = 0$, $c_1 + c_2 = s$ and $c_1/c_2 = \lambda_1/\lambda_2$, so $c_i = \lambda_i s / (\lambda_1 + \lambda_2)$ and $\theta = (\lambda_1 + \lambda_2)/s \leq \lambda_2$. This is consistent with the n_2 condition.

(c) By the envelope theorem, $\theta = \partial W^*/\partial s$. It is the marginal welfare value of one more unit of the *consumption good*, i.e. the shadow price of the good in welfare units. In the interior case it also equals the marginal disutility of labour (λ_2), because at the margin one more unit of good saves one unit of work.

(d) Make the good the numéraire ($p = 1$). Add a firm with technology $y = n$. Zero profit gives a wage $w = 1$ in goods. Property: person 1 owns the endowment s ; person 2 owns his labour. Person 1 makes a lump-sum transfer T to person 2 ($T < 0$ is a transfer from 2 to 1). Budgets:

$$c_1 \leq s - T, \quad c_2 \leq wn_2 + T.$$

Each person maximises his own utility: $\log c_1$ for person 1 and $\log c_2 - n_2$ for person 2.

(e) Person 1 consumes $c_1 = s - T$. Person 2 has FOC $w/c_2 = 1$, so $c_2 = w = 1$ and $n_2 = 1 - T$ (if positive). Goods market: $c_1 + c_2 = s - T + 1 = n_2 + s$. ✓ The equilibrium is $p = w = 1$, $c_1 = s - T$, $c_2 = 1$, $n_2 = 1 - T$. Without transfers ($T = 0$): $c_1 = s$, $c_2 = n_2 = 1$. That is the Pareto plan with $\lambda_1/\lambda_2 = s$. The transfer $T = s - \lambda_1/\lambda_2$ supports any interior Pareto plan of (b), as the second welfare theorem says. The budget multipliers are $\mu_1 = 1/c_1 = \lambda_2/\lambda_1$ and $\mu_2 = 1/c_2 = 1$, so $\mu_1/\mu_2 = \lambda_2/\lambda_1$, as in section 8.5.2.

(f) Pareto plan with fixed weights: in the interior region $dn_2/ds = -1$. Each extra unit of endowment is absorbed one-for-one by less work, and consumptions do not change, because utility is linear in n_2 (no income effect on consumption). Once $s \geq 1 + \lambda_1/\lambda_2$, $n_2 = 0$, and additional s raises both consumptions proportionally. In the competitive economy without transfers, $n_2 = 1$ for every s : the endowment belongs to person 1 and does not affect person 2's labour supply.

Numerical exercise — see the companion code

Illustration $\lambda_1 = 1$, $\lambda_2 = 2$. At $s = .5$: $(c_1, c_2, n_2, \theta) = (.5, 1, 1, 2)$; at $s = 3$ (corner): $(1, 2, 0, 1)$. `scipy`'s SLSQP on the planner reproduces both. A finite-difference $\partial W^*/\partial s$ equals θ . Person 2's problem at $w = 1$ with the transfer $T = s - \lambda_1/\lambda_2 = 0$ ($s = .5$) gives $n_2 = 1$.

Exercise 8.2 Family economics, II

Question

p. 305. Modify the economy in exercise 8.1 in the following way only. Instead of being endowed with s units of consumption, household 1 is endowed with 1 unit of the consumption good; and one unit of labor now produces sn_2 units of the consumption good, where $s > 0$. So a feasible allocation now satisfies

$$c_1 + c_2 \leq sn_2 + 1.$$

Preferences are identical with those described in exercise 8.25. Please answer counterparts of parts **a–f** for this family.

Solution:

Misprint. “Exercise 8.25” should be exercise 8.1 (8.25 is an asset-pricing exercise). We use the preferences of 8.1.

(a) $\mathcal{L} = \lambda_1 \log c_1 + \lambda_2 (\log c_2 - n_2) + \theta (sn_2 + 1 - c_1 - c_2)$, with $n_2 \geq 0$.

(b) FOCs: $\lambda_1/c_1 = \theta$, $\lambda_2/c_2 = \theta$, $-\lambda_2 + \theta s \leq 0$ ($= 0$ if $n_2 > 0$). Interior: $\theta = \lambda_2/s$, so $c_2 = \lambda_2/\theta = s$ and $c_1 = s\lambda_1/\lambda_2$. From the resource constraint, $n_2 = (c_1 + c_2 - 1)/s$:

$$c_1 = \frac{s\lambda_1}{\lambda_2}, \quad c_2 = s, \quad n_2 = 1 + \frac{\lambda_1}{\lambda_2} - \frac{1}{s}, \quad \theta = \frac{\lambda_2}{s},$$

valid when $s(1 + \lambda_1/\lambda_2) > 1$. Otherwise $n_2 = 0$, $c_i = \lambda_i/(\lambda_1 + \lambda_2)$, and $\theta = \lambda_1 + \lambda_2$ ($\geq \lambda_2/s$). ✓

(c) $\theta = \partial W^*/\partial(\text{endowment})$ is again the shadow price of the consumption good. Interior: $\theta = \lambda_2/s$ is the marginal disutility of labour per unit of good that labour produces.

(d) Good as numéraire; a firm with $y = sn$ pays wage $w = s$. Property: person 1 owns the unit endowment; person 2 owns his labour. Person 1 transfers T to person 2. Budgets: $c_1 \leq 1 - T$, $c_2 \leq wn_2 + T$.

(e) Person 2: FOC $w/c_2 = 1$, so $c_2 = w = s$ and $n_2 = (s - T)/s$. Person 1: $c_1 = 1 - T$. Goods market: $1 - T + s = sn_2 + 1$. ✓ With $T = 1 - s\lambda_1/\lambda_2$ this is the Pareto plan. Without transfers ($T = 0$): $c_1 = 1$, $c_2 = s$, $n_2 = 1$. That is the Pareto plan with $\lambda_1/\lambda_2 = 1/s$.

(f) In the Pareto plan (interior), $dn_2/ds = 1/s^2 > 0$: a more productive hour raises work. With no income effect on consumption ($c_2 = s$), the substitution effect alone operates. In the competitive economy without transfers, $n_2 = 1$ for every s : with log consumption, the income and substitution effects of the wage $w = s$ cancel.

Numerical exercise — see the companion code

Illustration $\lambda_1 = 1$, $\lambda_2 = 2$: $s = 2$ gives $(c_1, c_2, n_2, \theta) = (1, 2, 1, 1)$ (and $T = 0$); $s = .5$ gives the corner $(1/3, 2/3, 0, 3)$. SLSQP reproduces both. Person 2's problem at $w = s$, $T = 0$ gives $n_2 = 1$ for $s \in \{.5, 2, 7\}$.

Exercise 8.3 Existence of representative consumer

Question

p. 305. Suppose consumers 1 and 2 have one-period utility functions $u(c_1)$ and $w(c_2)$, respectively, where u and w are both increasing, strictly concave, twice differentiable functions of a scalar consumption rate. Let $c > 0$ be the total amount the single consumption good available to be allocated between consumers 1 and 2. Where $\theta \in (0, 1)$ is a Pareto weight, consider the Pareto problem:

$$v_\theta(c) = \max_{\{c_1, c_2\}} [\theta u(c_1) + (1 - \theta)w(c_2)]$$

subject to the constraint $c_1 + c_2 = c$. Show that the solution of this problem has the form of a concave utility function $v_\theta(c)$, which depends on the Pareto weight θ . Where $\{c_1(c, \theta), c_2(c, \theta)\}$ is a Pareto optimal allocation, show that $v'_\theta(c) = \theta u'(c_1(c, \theta)) = (1 - \theta)w'(c_2(c, \theta))$.

The function $v_\theta(c)$ is the utility function of a *representative consumer*. A representative consumer always lurks within a complete markets competitive equilibrium even with heterogeneous preferences. At a competitive equilibrium, the marginal utilities of the representative agent and each and every agent are proportional.

Solution:

Allocation. Substitute $c_2 = c - c_1$. The objective $\theta u(c_1) + (1 - \theta)w(c - c_1)$ is strictly concave in c_1 , so the maximiser is unique. When interior, it solves

$$\theta u'(c_1) = (1 - \theta)w'(c - c_1). \quad (1)$$

Derivative. Differentiate $v_\theta(c) = \theta u(c_1(c)) + (1 - \theta)w(c - c_1(c))$:

$$v'_\theta(c) = [\theta u'(c_1) - (1 - \theta)w'(c_2)] \frac{\partial c_1}{\partial c} + (1 - \theta)w'(c_2) = (1 - \theta)w'(c_2) = \theta u'(c_1),$$

where the bracket vanishes by (1) (envelope theorem).

Concavity. Differentiating (1) in c : $\theta u'' \partial c_1 / \partial c = (1 - \theta)w''(1 - \partial c_1 / \partial c)$, so

$$\frac{\partial c_1}{\partial c} = \frac{(1 - \theta)w''}{\theta u'' + (1 - \theta)w''} \in (0, 1), \quad v''_\theta(c) = \theta u''(c_1) \frac{\partial c_1}{\partial c} = \frac{\theta(1 - \theta)u''w''}{\theta u'' + (1 - \theta)w''} < 0,$$

since $u'', w'' < 0$ (both goods are normal). A derivative-free argument: take $c, \hat{c}$ with optimal allocations $(c_1, c_2), (\hat{c}_1, \hat{c}_2)$. For $\alpha \in (0, 1)$ the mix $\alpha(c_1, c_2) + (1 - \alpha)(\hat{c}_1, \hat{c}_2)$ is feasible for $\alpha c + (1 - \alpha)\hat{c}$. Concavity of u, w then gives $v_\theta(\alpha c + (1 - \alpha)\hat{c}) \geq \alpha v_\theta(c) + (1 - \alpha)v_\theta(\hat{c})$.

Representative consumer. In a complete-markets equilibrium, the allocation is Pareto optimal with $\theta/(1 - \theta) = \mu_2/\mu_1$ (section 8.5.2), and $c = c_1 + c_2$ is aggregate consumption at each (t, s^t) . Then $v'_\theta(c_t) = \theta u'(c_t^1) = (1 - \theta)w'(c_t^2)$, so the representative agent's marginal utility is proportional to each agent's. Hence prices $(*)$ can be computed from v_θ and the aggregate endowment alone.

Numerical exercise — see the companion code

Check with $u = \log$, $w(c) = -1/c$, $\theta = .3$: at $c = .5, 1, 2$, the finite-difference v'_θ equals $\theta u'(c_1) = (1 - \theta)w'(c_2)$ to 10^{-5} , and $v''_\theta < 0$ matches the formula above. E.g. $c = 1$: $c_1 = .24457$, $v' = 1.22663$.

Pricing in Endowment Economies (8.4–8.8)

Exercise 8.4 Term structure of interest rates

Question

p. 306. Consider an economy with a single consumer. There is one good in the economy, which arrives in the form of an exogenous endowment obeying

$$y_{t+1} = \lambda_{t+1}y_t,$$

where y_t is the endowment at time t and $\{\lambda_{t+1}\}$ is governed by a two-state Markov chain with transition matrix

$$P = \begin{bmatrix} p_{11} & 1 - p_{11} \\ 1 - p_{22} & p_{22} \end{bmatrix},$$

and initial distribution $\pi_\lambda = [\pi_0 \ 1 - \pi_0]$. The value of λ_t is given by $\bar{\lambda}_1 = .98$ in state 1 and $\bar{\lambda}_2 = 1.03$ in state 2. Assume that the history of y_s, λ_s up to t is observed at time t . The consumer has endowment process $\{y_t\}$ and has preferences over consumption streams that are ordered by

$$E_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$$

where $\beta \in (0, 1)$ and $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$, where $\gamma \geq 1$.

a. Define a competitive equilibrium, being careful to name all of the objects of which it consists.

b. Tell how to compute a competitive equilibrium.

For the remainder of this problem, suppose that $p_{11} = .8$, $p_{22} = .85$, $\pi_0 = .5$, $\beta = .96$, and $\gamma = 2$. Suppose that the economy begins with $\lambda_0 = .98$ and $y_0 = 1$.

c. Compute the (unconditional) average growth rate of consumption, computed before having observed λ_0 .

d. Compute the time 0 prices of three risk-free discount bonds, in particular, those promising to pay one unit of time j consumption for $j = 0, 1, 2$, respectively.

e. Compute the time 0 prices of three bonds, in particular, ones promising to pay one unit of time j consumption contingent on $\lambda_j = \bar{\lambda}_1$ for $j = 0, 1, 2$, respectively.

f. Compute the time 0 prices of three bonds, in particular, ones promising to pay one unit of time j consumption contingent on $\lambda_j = \bar{\lambda}_2$ for $j = 0, 1, 2$, respectively.

g. Compare the prices that you computed in parts d, e, and f.

Solution:

(a) The state is $s_t = \lambda_t$; a history is $s^t = (\lambda_0, \dots, \lambda_t)$ and $y_t(s^t) = y_0 \prod_{k=1}^t \lambda_k$. A competitive equilibrium (time-0 trading) is a *price system* $\{q_t^0(s^t)\}$ and an *allocation* $\{c_t(s^t)\}$ such that: (i) given prices, $\{c_t\}$ maximises $\sum_t \sum_{s^t} \beta^t u(c_t(s^t)) \pi_t(s^t)$ subject to $\sum_t \sum_{s^t} q_t^0(s^t) c_t(s^t) \leq \sum_t \sum_{s^t} q_t^0(s^t) y_t(s^t)$; (ii) the allocation is feasible, $c_t(s^t) \leq y_t(s^t)$.

(b) With one consumer, feasibility and non-satiation force $c_t(s^t) = y_t(s^t)$. Prices then

come from the consumer's FOC (*) evaluated at $c = y$:

$$q_t^0(s^t) = \beta^t \pi_t(s^t) \left(\frac{y_t}{y_0} \right)^{-\gamma} = \beta^t \pi_t(s^t) \prod_{k=1}^t \lambda_k^{-\gamma}.$$

In matrix form, let $Q_{ij} = \beta P_{ij} \bar{\lambda}_j^{-\gamma}$ (price in units of today's good of one unit of tomorrow's good if tomorrow's state is j). Then the time-0 price of one unit at t in state j is $(Q^t)_{\lambda_0, j}$. So the price of any claim follows by summing.

(c) The stationary distribution solves $\bar{\pi} = \bar{\pi}P$: $.2\bar{\pi}_1 = .15\bar{\pi}_2$, so $\bar{\pi} = (3/7, 4/7)$. The unconditional mean gross growth rate is

$$E[\lambda] = \frac{3}{7}(.98) + \frac{4}{7}(1.03) = \frac{7.06}{7} = 1.008571,$$

i.e. 0.857% per period. *Reading.* If instead one averages with the given initial distribution $\pi_\lambda = (.5, .5)$, then $E\lambda_0 = 1.005$, $E\lambda_1 = \pi_\lambda P \bar{\lambda} = 1.00625$, and $E\lambda_t = \pi_\lambda P^t \bar{\lambda} \rightarrow 1.008571$. The ergodic mean is the natural “unconditional” answer.

(d)–(f) With $\lambda_0 = \bar{\lambda}_1$ (row 1):

$$Q = .96 \begin{bmatrix} .8(.98)^{-2} & .2(1.03)^{-2} \\ .15(.98)^{-2} & .85(1.03)^{-2} \end{bmatrix} = \begin{bmatrix} .799667 & .180978 \\ .149938 & .769158 \end{bmatrix}.$$

Risk-free price = $e_1' Q^j \mathbf{1}$; contingent on $\bar{\lambda}_1$: $(Q^j)_{11}$; on $\bar{\lambda}_2$: $(Q^j)_{12}$.

j	(d) risk-free	(e) if $\lambda_j = .98$	(f) if $\lambda_j = 1.03$
0	1	1	0
1	.980645	.799667	.180978
2	.950526	.666602	.283923

At $j = 0$ the state $\lambda_0 = .98$ is known, so the claim contingent on $\bar{\lambda}_1$ is worth one unit and the one contingent on $\bar{\lambda}_2$ is worthless.

(g) (d) = (e) + (f) for each j : a risk-free bond is a portfolio of the two contingent claims (no arbitrage). Yields $p_j^{-1/j} - 1$ are 1.974% ($j = 1$) and 2.569% per period ($j = 2$): the yield curve slopes up. The economy starts in the low-growth state, which is persistent. Expected consumption growth, and hence the interest rate, is low in the near term and rises towards the ergodic mean. Contingent prices divided by probabilities exceed β^j in the low-growth state: marginal utility is high when growth is low.

Numerical exercise — see the companion code

All nine prices are checked by enumerating every history of length j and summing $\beta^j \pi(\text{history}) \prod \lambda^{-\gamma}$.

Exercise 8.5 Periodic endowments

Question

p. 307. An economy consists of two infinitely lived consumers named $i = 1, 2$. There is one nonstorable consumption good. Consumer i consumes c_t^i at time t . Consumer i ranks consumption streams by

$$\sum_{t=0}^{\infty} \beta^t u(c_t^i),$$

where $\beta \in (0, 1)$ and $u(c)$ is increasing, strictly concave, and twice continuously differentiable. Consumer 1 is endowed with a stream of the consumption good $y_t^1 = 1, 0, 0, 1, 0, 0, 1, \dots$. Consumer 2 is endowed with a stream of the consumption good $0, 1, 1, 0, 1, 1, 0, \dots$. Assume that there are complete markets with time 0 trading.

- a. Define a competitive equilibrium.
- b. Compute a competitive equilibrium.
- c. Suppose that one of the consumers markets a derivative asset that promises to pay .05 units of consumption each period. What would the price of that asset be?

Solution:

(a) A price sequence $\{q_t^0\}$ and allocations $\{c_t^1, c_t^2\}$ such that each c^i maximises $\sum_t \beta^t u(c_t^i)$ s.t. $\sum_t q_t^0 c_t^i \leq \sum_t q_t^0 y_t^i$, and $c_t^1 + c_t^2 = y_t^1 + y_t^2$ for all t .

(b) The aggregate endowment is $y_t^1 + y_t^2 = 1$ for every t . FOC (8.5.4): $\beta^t u'(c_t^i) = \mu_i q_t^0$, so $u'(c_t^1)/u'(c_t^2) = \mu_1/\mu_2$ for all t . With $c_t^2 = 1 - c_t^1$, the ratio $u'(c)/u'(1 - c)$ is strictly decreasing in c , so $c_t^1 = c^1$ is constant. Then $q_t^0 = \beta^t u'(c^1)/\mu_1$, and the normalisation $q_0^0 = 1$ gives

$$q_t^0 = \beta^t.$$

Consumer 1's budget: $c^1 \sum_t \beta^t = \sum_{k \geq 0} \beta^{3k}$, i.e. $c^1/(1 - \beta) = 1/(1 - \beta^3)$, so

$$c_t^1 = \frac{1 - \beta}{1 - \beta^3} = \frac{1}{1 + \beta + \beta^2}, \quad c_t^2 = \frac{\beta + \beta^2}{1 + \beta + \beta^2}.$$

Consumption shares equal wealth shares. Consumer 1 owns one-third of the dates, but his dates come first: his wealth $1/(1 - \beta^3)$ exceeds consumer 2's $(\beta + \beta^2)/(1 - \beta^3)$ iff $\beta + \beta^2 < 1$, i.e. $\beta < .618$.

(c) The asset is a riskless consol paying .05 at every $t \geq 0$. By (8.7.1)–(8.7.2):

$$p = .05 \sum_{t \geq 0} q_t^0 = \frac{.05}{1 - \beta} \quad \left(\text{ex-dividend: } \frac{.05\beta}{1 - \beta} \right).$$

Numerical exercise — see the companion code

Illustration $\beta = .95$: $c^1 = .350570$, $c^2 = .649430$, $p = 1$ (.95 ex-dividend). Both budgets are checked by truncated sums ($T = 3000$).

Exercise 8.6 Mehra–Prescott endowment, two market structures

Question

p. 307. Consider a pure endowment economy with a single representative consumer; $\{c_t, d_t\}_{t=0}^{\infty}$ are the consumption and endowment processes, respectively. Feasible allocations satisfy

$$c_t \leq d_t.$$

The endowment process is described by

$$d_{t+1} = \lambda_{t+1} d_t.$$

The growth rate λ_{t+1} is described by a two-state Markov process with transition probabilities

$$P_{ij} = \text{Prob}(\lambda_{t+1} = \bar{\lambda}_j | \lambda_t = \bar{\lambda}_i).$$

Assume that

$$P = \begin{bmatrix} .8 & .2 \\ .1 & .9 \end{bmatrix}, \quad \text{and that} \quad \bar{\lambda} = \begin{bmatrix} .97 \\ 1.03 \end{bmatrix}.$$

In addition, $\lambda_0 = .97$ and $d_0 = 1$ are both known at date 0. The consumer has preferences over consumption ordered by

$$E_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\gamma}}{1-\gamma},$$

where E_0 is the mathematical expectation operator, conditioned on information known at time 0, $\gamma = 2$, $\beta = .95$.

Part I. At time 0, after d_0 and λ_0 are known, there are complete markets in date- and history-contingent claims. The market prices are denominated in units of time 0 consumption goods.

a. Define a competitive equilibrium, being careful to specify all the objects composing an equilibrium.

b. Compute the equilibrium price of a claim to one unit of consumption at date 5, denominated in units of time 0 consumption, contingent on the following history of growth rates: $(\lambda_1, \lambda_2, \dots, \lambda_5) = (.97, .97, 1.03, .97, 1.03)$. Please give a numerical answer.

c. Compute the equilibrium price of a claim to one unit of consumption at date 5, denominated in units of time 0 consumption, contingent on the following history of growth rates: $(\lambda_1, \lambda_2, \dots, \lambda_5) = (1.03, 1.03, 1.03, 1.03, .97)$.

d. Give a formula for the price at time 0 of a claim on the entire endowment sequence.

e. Give a formula for the price at time 0 of a claim on consumption in period 5, contingent on the growth rate λ_5 being .97 (regardless of the intervening growth rates).

Part II. Now assume a different market structure. Assume that at each date $t \geq 0$ there is a complete set of one-period forward Arrow securities.

f. Define a (recursive) competitive equilibrium with Arrow securities, being careful to define all of the objects that compose such an equilibrium.

g. For the representative consumer in this economy, for each state compute the “natural debt limits” that constrain state-contingent borrowing.

h. Compute a competitive equilibrium with Arrow securities. In particular, compute both the pricing kernel and the allocation.

Solution:

Let state 1 be $\lambda = .97$ and state 2 be $\lambda = 1.03$. Since $u'(c) = c^{-\gamma}$ and $c = d$, $u'(d_{t+1})/u'(d_t) = \lambda_{t+1}^{-\gamma}$.

(a) A price system $\{q_t^0(\lambda^t)\}$ and an allocation $\{c_t(\lambda^t)\}$ such that c maximises $E_0 \sum \beta^t c_t^{1-\gamma}/(1-\gamma)$ s.t. $\sum_t \sum_{\lambda^t} q_t^0 c_t \leq \sum_t \sum_{\lambda^t} q_t^0 d_t$, and $c_t(\lambda^t) = d_t(\lambda^t)$ (feasibility with one consumer). From (*): $q_t^0(\lambda^t) = \beta^t \pi_t(\lambda^t)(d_t/d_0)^{-\gamma}$.

(b) The path $.97 \rightarrow .97 \rightarrow .97 \rightarrow 1.03 \rightarrow .97 \rightarrow 1.03$ has probability $.8 \cdot .8 \cdot .2 \cdot .1 \cdot .2 = .00256$. Also $d_5 = .97^3 1.03^2 = .968255$. So

$$q_5^0 = .95^5 (.00256) (.968255)^{-2} = \mathbf{.00211290}.$$

(c) Probability $.2 \cdot .9^3 \cdot .1 = .01458$; $d_5 = 1.03^4 (.97) = 1.091744$; so

$$q_5^0 = .95^5 (.01458) (1.091744)^{-2} = \mathbf{.00946530}.$$

The price per unit of probability is lower in (c): the high-growth path has high d_5 and low marginal utility.

(d) Let $K_{ij} = \beta P_{ij} \bar{\lambda}_j^{1-\gamma}$. Then $\sum_{\lambda^t} q_t^0 d_t = d_0 e'_{\lambda_0} K^t \mathbf{1}$, and

$$p_0^0 = \sum_{t \geq 0} \sum_{\lambda^t} q_t^0(\lambda^t) d_t(\lambda^t) = d_0 e'_{\lambda_0} (I - K)^{-1} \mathbf{1} \equiv d_0 v(\lambda_0),$$

provided the spectral radius of K is below one. Numerically $v = (18.9332, 16.7995)$, so $p_0^0 = 18.9332$ (cum-dividend; 17.9332 ex-dividend).

(e) With $Q_{ij} = \beta P_{ij} \bar{\lambda}_j^{-\gamma}$, one unit at $t = 5$ if $\lambda_5 = .97$ costs $(Q^5)_{11} = .440235$. A claim to the date-5 endowment d_5 in that event costs $d_0 (K^5)_{11} = .388790$.

(f) Let the state be (a, d, λ) , with a the Arrow-security claims brought into the period. Objects: a pricing kernel $Q(\lambda'|\lambda)$ (price today of one unit tomorrow if tomorrow's growth is λ'); debt limits $A(d', \lambda')$; a value function $v(a, d, \lambda)$; decision rules $c = h(a, d, \lambda)$, $\hat{a}(\lambda') = g(a, d, \lambda, \lambda')$; initial wealth $a_0 = 0$. They must satisfy: (i) given Q and A , v, h, g solve

$$v(a, d, \lambda) = \max_{c, \hat{a}(\cdot)} \left\{ \frac{c^{1-\gamma}}{1-\gamma} + \beta \sum_{\lambda'} P(\lambda'|\lambda) v(\hat{a}(\lambda'), \lambda' d, \lambda') \right\}$$

s.t. $c + \sum_{\lambda'} Q(\lambda'|\lambda) \hat{a}(\lambda') \leq d + a$ and $-\hat{a}(\lambda') \leq A(\lambda' d, \lambda')$; (ii) markets clear: $c = d$ and $\hat{a}(\lambda') = 0$ (zero net supply, one consumer).

(g) The natural debt limit is the value of the endowment tail, (8.8.2):

$$A(d_t, \lambda_t) = \sum_{\tau \geq t} \sum_{\lambda^\tau | \lambda^t} q_\tau^t d_\tau = d_t v(\lambda_t) = \begin{cases} 18.9332 d_t, & \lambda_t = .97, \\ 16.7995 d_t, & \lambda_t = 1.03. \end{cases}$$

Seen from date t , the limit on claims for state λ' at $t + 1$ is $d_t \lambda' v(\lambda')$. It satisfies the recursion $v = \mathbf{1} + K v$, i.e. (8.9.13) scaled by d .

(h) $c_t = d_t$, $\hat{a} \equiv 0$, and from (8.9.12)

$$Q(\bar{\lambda}_j | \bar{\lambda}_i) = \beta P_{ij} \bar{\lambda}_j^{-\gamma} : \quad Q = \begin{bmatrix} .807737 & .179093 \\ .100967 & .805920 \end{bmatrix}.$$

(i) *Reading.* The bond is called “two-period” but pays at $t + 1$; we price the stated payoff, a sure unit at $t + 1$. Its price is the row sum of Q :

$$p^b(\lambda_t) = \beta \sum_j P_{ij} \bar{\lambda}_j^{-\gamma} = \begin{cases} .986830, & \lambda_t = .97, \\ .906887, & \lambda_t = 1.03. \end{cases}$$

If the bond pays at $t + 2$ instead, its price is the row sum of Q^2 : .959517 and .830515.

Numerical exercise — see the companion code

(b), (c) and (e) are checked by enumerating all 2^5 histories. For v , the inverse is checked against the truncated series $\sum_{j < 4000} K^j \mathbf{1}$, and the recursion $v = \mathbf{1} + Kv$ is verified.

Exercise 8.7 Two consumers, Markov endowment

Question

pp. 309–311. An economy consists of two consumers, named $i = 1, 2$. The economy exists in discrete time for periods $t \geq 0$. There is one good in the economy, which is not storable and arrives in the form of an endowment stream owned by each consumer. The endowments to consumers $i = 1, 2$ are

$$y_t^1 = s_t, \quad y_t^2 = 1$$

where s_t is a random variable governed by a two-state Markov chain with values $s_t = \bar{s}_1 = 0$ or $s_t = \bar{s}_2 = 1$. The Markov chain has time invariant transition probabilities denoted by $\pi(s_{t+1} = s' | s_t = s) = \pi(s' | s)$, and the probability distribution over the initial state is $\pi_0(s)$. The *aggregate endowment* at t is $Y(s_t) = y_t^1 + y_t^2$. Let c^i denote the stochastic process of consumption for agent i . Consumer i orders consumption streams according to

$$U(c^i) = \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \ln[c_t^i(s^t)] \pi_t(s^t),$$

where $\pi_t(s^t)$ is the probability of the history $s^t = (s_0, s_1, \dots, s_t)$.

- a. Give a formula for $\pi_t(s^t)$.
- b. Let $\theta \in (0, 1)$ be a Pareto weight on consumer 1. Consider the planning problem $\max_{c^1, c^2} \{\theta U(c^1) + (1 - \theta)U(c^2)\}$ where maximization is subject to $c_t^1(s^t) + c_t^2(s^t) \leq Y(s_t)$, $\forall t, \forall s^t$. Solve the Pareto problem, taking θ as a parameter.
- c. Define a *competitive equilibrium* with history-dependent Arrow-Debreu securities traded once and for all at time 0. Be careful to define all of the objects that compose a competitive equilibrium.
- d. Compute the competitive equilibrium price system (i.e., find the prices of all Arrow-Debreu securities).
- e. Tell the relationship between the solutions (indexed by θ) of the Pareto problem and the competitive equilibrium allocation. If you wish, refer to the two welfare theorems.
- f. Briefly tell how you can compute the competitive equilibrium price system *before* you have figured out the competitive equilibrium allocation.
- g. Now define a recursive competitive equilibrium with trading every period in one-period Arrow securities only. Describe all of the objects of which such an equilibrium is composed. (Please denominate the prices of one-period time $t + 1$ state-contingent Arrow securities in units of time t consumption.) Define “natural borrowing limits” for each consumer in each state. Tell how to compute these natural borrowing limits.
- h. Tell how to compute the prices of one-period Arrow securities. How many prices are there (i.e., how many numbers do you have to compute)? Compute all of these prices in the special case that $\beta = .95$ and $\pi(s_j | s_i) = P_{ij}$ where $P = \begin{bmatrix} .8 & .2 \\ .3 & .7 \end{bmatrix}$.
- i. Within the one-period Arrow securities economy, a new asset is introduced. One of the consumers decides to market a one-period-ahead riskless claim to one unit of consumption. Compute the equilibrium price of this security when $s_t = 0$ and when $s_t = 1$. Justify your formula for these prices in terms of first principles.
- j. Within the one-period Arrow securities equilibrium, a new asset is introduced. One of the consumers decides to market a two-period-ahead riskless claim to one unit of consumption (a two-period real bill). Compute the equilibrium prices of this

Solution:

(a) By the Markov property (8.9.1): $\pi_t(s^t) = \pi_0(s_0) \pi(s_1|s_0) \pi(s_2|s_1) \cdots \pi(s_t|s_{t-1})$. Since trade starts after s_0 is seen, $\pi_0(s_0) = 1$ for the realised s_0 .

(b) Lagrangian with multipliers $\beta^t \pi_t(s^t) \zeta_t(s^t)$ on the resource constraints. FOCs: $\theta/c_t^1(s^t) = \zeta_t(s^t) = (1 - \theta)/c_t^2(s^t)$, so $c_t^1/c_t^2 = \theta/(1 - \theta)$. With $c_t^1 + c_t^2 = Y(s_t)$:

$$\boxed{c_t^1(s^t) = \theta Y(s_t), \quad c_t^2(s^t) = (1 - \theta) Y(s_t).}$$

Consumption depends only on the current aggregate endowment (Proposition 1).

(c) A price system $\{q_t^0(s^t)\}$ and an allocation $\{c_t^1(s^t), c_t^2(s^t)\}$ such that: (i) for each i , c^i maximises $U(c^i)$ s.t. $\sum_t \sum_{s^t} q_t^0(s^t) c_t^i(s^t) \leq \sum_t \sum_{s^t} q_t^0(s^t) y_t^i(s^t)$; (ii) $c_t^1(s^t) + c_t^2(s^t) = Y(s_t)$ for all t, s^t .

(d) FOC (8.5.4): $\beta^t \pi_t(s^t)/c_t^i(s^t) = \mu_i q_t^0(s^t)$. With $c^1 = \theta Y$ and $q_0^0(s_0) = 1$, we get $\mu_1 = 1/(\theta Y(s_0))$, so

$$\boxed{q_t^0(s^t) = \beta^t \pi_t(s^t) \frac{Y(s_0)}{Y(s_t)}}.$$

θ is pinned down by consumer 1's budget. The value of his consumption is $\sum_t \sum_{s^t} \beta^t \pi_t \theta Y(s_0) = \theta Y(s_0)/(1 - \beta)$. The value of his endowment is $Y(s_0) \sum_t \beta^t E_{s_0}[y^1(s_t)/Y(s_t)]$. Hence

$$\theta = (1 - \beta) \sum_{t \geq 0} \beta^t E_{s_0} \left[\frac{y^1(s_t)}{Y(s_t)} \right] = (1 - \beta) e'_{s_0} (I - \beta P)^{-1} \left(\frac{y^1}{Y} \right).$$

Here $y^1/Y = (0, \frac{1}{2})$. With the numbers of (h): $\theta = .180952$ if $s_0 = 0$, and $\theta = .228571$ if $s_0 = 1$.

(e) First welfare theorem: the equilibrium allocation is the Pareto allocation of (b) with $\theta/(1 - \theta) = \mu_2/\mu_1$, i.e. θ equal to consumer 1's wealth share above. Second welfare theorem: every $\theta \in (0, 1)$ is a competitive allocation for a suitable redistribution of initial wealth; the prices in (d) are the same for every θ .

(f) Because the allocation is $c^i = \theta_i Y$ for *some* θ_i and utility is logarithmic, the marginal rate of substitution $u'(\theta Y_t)/u'(\theta Y_0) = Y_0/Y_t$ does not depend on θ . So (*) gives q_t^0 from the aggregate endowment alone, before θ is found. The same holds for any identical CRRA preferences (a representative consumer with $u(Y)$).

(g) Objects: a pricing kernel $Q(s'|s)$ (price at t , in time- t goods, of one unit at $t + 1$ if $s_{t+1} = s'$); borrowing limits $\bar{A}^i(s)$; value functions $v^i(a, s)$; decision rules $c = h^i(a, s)$, $\hat{a}(s') = g^i(a, s, s')$; initial wealth $a_0^i = 0$. Consumer i solves

$$\begin{aligned} v^i(a, s) &= \max_{c, \hat{a}(\cdot)} \left\{ \ln c + \beta \sum_{s'} v^i(\hat{a}(s'), s') \pi(s'|s) \right\} \\ \text{s.t. } & c + \sum_{s'} Q(s'|s) \hat{a}(s') \leq y^i(s) + a, \quad -\hat{a}(s') \leq \bar{A}^i(s'). \end{aligned}$$

Markets clear: $\sum_i c^i = Y(s)$ and $\sum_i \hat{a}^i(s') = 0$. *Natural borrowing limits* are the most a consumer can repay from his own endowment if he consumes nothing from then on

(8.8.2). They solve the linear system (8.9.13):

$$\bar{A}^i(s) = y^i(s) + \sum_{s'} Q(s'|s) \bar{A}^i(s') \iff \bar{A}^i = (I - Q)^{-1} y^i.$$

(h) From (8.9.12) with $c^i = \theta_i Y$: $Q(s'|s) = \beta \pi(s'|s) Y(s)/Y(s')$. There are $2 \times 2 = 4$ prices. With $Y = (1, 2)$:

$$Q = \begin{bmatrix} .95(.8)(1/1) & .95(.2)(1/2) \\ .95(.3)(2/1) & .95(.7)(2/2) \end{bmatrix} = \begin{bmatrix} .760 & .095 \\ .570 & .665 \end{bmatrix}.$$

Natural borrowing limits: $\bar{A}^1 = (3.6190, 9.1429)$ and $\bar{A}^2 = (16.3810, 30.8571)$. Equilibrium holdings are $\hat{a}^1(s') = \Upsilon^1(s') = [(I - Q)^{-1}(c^1 - y^1)](s')$. For $s_0 = 0$: $\Upsilon^1 = (0, -1.9048)$; consumer 1 enters state 1 in debt because his endowment is high there.

(i) A sure unit next period is the bundle of one Arrow security for each s' . Absence of arbitrage (or either consumer's Euler equation $p = \beta E[u'(c')/u'(c) | s]$) gives $p(s) = \sum_{s'} Q(s'|s)$:

$$p(0) = .760 + .095 = .855, \quad p(1) = .570 + .665 = 1.235.$$

In state 1 the endowment is expected to fall ($\pi(0|1) = .3$), so consumers want to save and the gross risk-free rate is $1/1.235 = .81 < 1$.

(j) By (8.10.3), $Q_2 = Q^2$, and the bill costs $e'_s Q^2 \mathbf{1}$: .767125 ($s = 0$) and 1.308625 ($s = 1$).

(k) $Q_5 = Q^5$:

$$Q^5 = \begin{bmatrix} .473941 & .149920 \\ .899520 & .324021 \end{bmatrix} \quad (\text{row } s_t, \text{ column } s_{t+5}).$$

Numerical exercise — see the companion code

Q^2 and Q^5 from the explicit double sum (8.10.3) equal the matrix powers. θ from the inverse equals a truncated sum of $\beta^t P^t$. With $a = \Upsilon$, the budget $c + \sum_{s'} Q(s'|s) \Upsilon(s') = y(s) + \Upsilon(s)$ holds in both states, and $\Upsilon^1(s_0) = 0$.

Exercise 8.8 Optimal taxation

Question

pp. 311–312. The government of a small country must finance an exogenous stream of government purchases $\{g_t\}_{t=0}^{\infty}$. Assume that g_t is described by a discrete-state Markov chain with transition matrix P and initial distribution π_0 . Let $\pi_t(g^t)$ denote the probability of history $g^t = g_t, g_{t-1}, \dots, g_0$, conditioned on g_0 . The state of the economy is completely described by the history g^t . There are complete markets in date-history claims to goods. At time 0, after g_0 has been realized, the government can purchase or sell claims to time t goods contingent on history g^t at a price $p_t^0(g^t) = \beta^t \pi_t(g^t)$, where $\beta \in (0, 1)$. The date-state prices are exogenous to the small country. The government finances its expenditures by raising history-contingent tax revenues of $R_t = R_t(g^t)$ at time t . The present value of its expenditures must not exceed the value of its revenues, where values are calculated using Arrow-Debreu prices to evaluate state-contingent streams of consumption goods.

Raising revenues by taxation is distorting. The government confronts a dead weight loss function $W(R_t)$ that measures the distortion at time t . Assume that W is an increasing, twice differentiable, strictly convex function that satisfies $W(0) = 0$, $W'(0) = 0$, $W'(R) > 0$ for $R > 0$ and $W''(R) > 0$ for $R \geq 0$. The government devises a state-contingent taxation and borrowing plan to minimize

$$E_0 \sum_{t=0}^{\infty} \beta^t W(R_t), \quad (1)$$

where E_0 is the mathematical expectation conditioned on g_0 .

Suppose that g_t takes two possible values, $\bar{g}_1 = .2$ (peace) and $\bar{g}_2 = 1$ (war) and that $P = \begin{bmatrix} .8 & .2 \\ .5 & .5 \end{bmatrix}$. Suppose that $g_0 = .2$. Finally, suppose that $W(R) = .5R^2$.

- a. Please write out (1) long hand, i.e., write out an explicit expression for the mathematical expectation E_0 in terms of a summation over the appropriate probability distribution.
- b. Compute the optimal tax and borrowing plan. In particular, give analytic expressions for $R_t = R_t(g^t)$ for all t and all g^t .
- c. There is an equivalent market setting in which the government can buy and sell one-period Arrow securities each period. Find the price of one-period Arrow securities at time t , denominated in units of the time t good.
- d. Let $B_t(g_t)$ be the one-period Arrow securities at t that the government issued for state g_t at time $t - 1$. For $t > 0$, compute $B_t(g_t)$ for $g_t = \bar{g}_1$ and $g_t = \bar{g}_2$.
- e. Use your answers to parts b and d to describe the government's optimal policy for taxing and borrowing.

Solution:

(a)

$$E_0 \sum_{t \geq 0} \beta^t W(R_t) = \sum_{t=0}^{\infty} \sum_{g^t} \beta^t \pi_t(g^t) .5 R_t(g^t)^2, \quad \pi_t(g^t) = \prod_{k=1}^t P(g_k | g_{k-1}), \quad g_0 = .2.$$

(b) Lagrangian with multiplier μ on the budget $\sum_t \sum_{g^t} \beta^t \pi_t(g^t) [R_t(g^t) - g_t] \geq 0$:

$$\mathcal{L} = \sum_t \sum_{g^t} \beta^t \pi_t(g^t) \{W(R_t) - \mu [R_t - g_t]\}.$$

FOC: $\beta^t \pi_t W'(R_t) = \mu \beta^t \pi_t$, so $W'(R_t) = R_t = \mu$ for all t, g^t . Revenue is **constant**: $R_t(g^t) = R$. The budget gives $R/(1 - \beta) = \sum_t \beta^t E_0 g_t = e'_1 (I - \beta P)^{-1} \bar{g} \equiv v_1$. Write $v = (I - \beta P)^{-1} \bar{g}$:

$$v_1 = .2 + \beta(.8v_1 + .2v_2), \quad v_2 = 1 + \beta(.5v_1 + .5v_2).$$

The second gives $v_2 = (1 + .5\beta v_1)/(1 - .5\beta)$. Substituting into the first: $v_1[(1 - .8\beta)(1 - .5\beta) - .1\beta^2] = .2(1 - .5\beta) + .2\beta$. The bracket is $1 - 1.3\beta + .3\beta^2 = (1 - \beta)(1 - .3\beta)$, so

$$v_1 = \frac{.2 + .1\beta}{(1 - \beta)(1 - .3\beta)}, \quad \boxed{R_t(g^t) = R = (1 - \beta)v_1 = \frac{.2 + .1\beta}{1 - .3\beta} \quad \forall t, g^t.}$$

Checks: $\beta \rightarrow 0$ gives $R = .2 = g_0$; $\beta \rightarrow 1$ gives $R = 3/7$, the stationary mean of g ($\bar{\pi} = (5/7, 2/7)$).

(c) $Q(g_{t+1}|g_t) = p_{t+1}^0(g^{t+1})/p_t^0(g^t) = \beta P(g_{t+1}|g_t)$, i.e. $Q = \beta P = \begin{bmatrix} .8\beta & .2\beta \\ .5\beta & .5\beta \end{bmatrix}$.

(d) Sequential budget: $g_t + B_t(g_t) = R + \sum_{g'} Q(g'|g_t) B_{t+1}(g')$. Iterating forward (no Ponzi), the debt due equals the present value of future surpluses:

$$B_t(g_t) = \sum_{j \geq 0} \beta^j E_t[R - g_{t+j}] = \frac{R}{1 - \beta} - v(g_t).$$

State 1: $B(\bar{g}_1) = R/(1 - \beta) - v_1 = 0$. State 2: from $v_2(1 - .5\beta) = 1 + .5\beta v_1$,

$$v_2 - v_1 = \frac{1 - (1 - \beta)v_1}{1 - .5\beta} = \frac{1 - R}{1 - .5\beta}, \quad 1 - R = \frac{.8(1 - .5\beta)}{1 - .3\beta},$$

so

$$\boxed{B_t(\bar{g}_1) = 0, \quad B_t(\bar{g}_2) = -\frac{.8}{1 - .3\beta}.$$

(e) Tax revenue is perfectly smoothed across dates and states: W is convex and prices are actuarially fair ($p_t^0 = \beta^t \pi_t$). The government uses Arrow securities as *war insurance*. In peace it runs a surplus $R - .2$ and spends it buying claims that pay $.8/(1 - .3\beta)$ if war breaks out next period: $Q(\bar{g}_2|\bar{g}_1) \cdot .8/(1 - .3\beta) = .2\beta \cdot .8/(1 - .3\beta) = R - .2$. It never owes anything when peace comes ($B(\bar{g}_1) = 0$). In war it collects the insurance, pays

the excess of g over R , and buys insurance again for next period. The position depends only on g_t , not on the history.

Numerical exercise — see the companion code

Illustration $\beta = .95$ (the book leaves β free): $R = .295/.715 = .412587$, $B(\bar{g}_2) = -1.118881$. `scipy` minimisation over state-contingent $(R(\bar{g}_1), R(\bar{g}_2))$ returns R in both states. The sequential budget holds in both states.

Linear and Logarithmic Consumers: Corners (8.9–8.10)

Exercise 8.9 A competitive equilibrium

Question

pp. 312–313. An endowment economy consists of two type of consumers. Consumers of type 1 order consumption streams of the one good according to

$$\sum_{t=0}^{\infty} \beta^t c_t^1$$

and consumers of type 2 order consumption streams according to

$$\sum_{t=0}^{\infty} \beta^t \ln(c_t^2)$$

where $c_t^i \geq 0$ is the consumption of a type i consumer and $\beta \in (0, 1)$ is a common discount factor. The consumption good is tradable but nonstorable. There are equal numbers of the two types of consumer. The consumer of type 1 is endowed with the consumption sequence

$$y_t^1 = \mu > 0 \quad \forall t \geq 0$$

where $\mu > 0$. The consumer of type 2 is endowed with the consumption sequence

$$y_t^2 = \begin{cases} 0 & \text{if } t \geq 0 \text{ is even} \\ \alpha & \text{if } t \geq 0 \text{ is odd} \end{cases}$$

where $\alpha = \mu(1 + \beta^{-1})$.

- Define a competitive equilibrium with time 0 trading. Be careful to include definitions of all of the objects of which a competitive equilibrium is composed.
- Compute a competitive equilibrium allocation with time 0 trading.
- Compute the time 0 wealths of the two types of consumers using the competitive equilibrium prices.
- Define a competitive equilibrium with sequential trading of Arrow securities.
- Compute a competitive equilibrium with sequential trading of Arrow securities.

Solution:

(a) A price sequence $\{q_t^0\}$ and allocations $\{c_t^1, c_t^2\}$ such that each type maximises its utility subject to $\sum_t q_t^0 c_t^i \leq \sum_t q_t^0 y_t^i$ and $c_t^i \geq 0$, and $c_t^1 + c_t^2 = y_t^1 + y_t^2$ for all t .

(b) Kuhn–Tucker conditions with budget multipliers μ_1, μ_2 :

$$\text{type 1: } \beta^t \leq \mu_1 q_t^0 \quad (= \text{ if } c_t^1 > 0); \quad \text{type 2: } \beta^t / c_t^2 = \mu_2 q_t^0.$$

Guess $q_t^0 = \beta^t$ (type 1 interior, $\mu_1 = 1$). Then $c_t^2 = 1/\mu_2 \equiv \bar{c}$ is constant. Type 2's budget:

$$\frac{\bar{c}}{1 - \beta} = \alpha(\beta + \beta^3 + \cdots) = \frac{\alpha\beta}{1 - \beta^2} \implies \bar{c} = \frac{\alpha\beta}{1 + \beta} = \frac{\mu(1 + \beta^{-1})\beta}{1 + \beta} = \mu.$$

Feasibility: $c_t^1 = \mu + y_t^2 - \mu = y_t^2$. So

$$q_t^0 = \beta^t, \quad c_t^2 = \mu \quad \forall t, \quad c_t^1 = \begin{cases} 0, & t \text{ even,} \\ \alpha, & t \text{ odd.} \end{cases}$$

$c_t^1 \geq 0$ holds, and at even dates type 1 is at the corner $c_t^1 = 0$ with the KKT condition holding with equality. The value $\alpha = \mu(1 + \beta^{-1})$ is exactly the knife edge at which type 1's non-negativity constraint is just reached.

(c) Type 1: $\sum_t \beta^t \mu = \mu/(1 - \beta)$. Type 2: $\alpha\beta/(1 - \beta^2) = \mu/(1 - \beta)$. Wealths are **equal**.

(d) No uncertainty, so the Arrow security is a one-period bond. At each t it has price Q_t (time- t goods per unit at $t + 1$). Consumer i faces $c_t^i + Q_t a_{t+1}^i \leq y_t^i + a_t^i$, $c_t^i \geq 0$, $-a_{t+1}^i \leq A_{t+1}^i$, $a_0^i = 0$, with natural debt limits $A_t^i = \sum_{\tau \geq t} (q_\tau^0/q_t^0) y_\tau^i$. An equilibrium is $\{Q_t\}$, $\{A_t^i\}$, allocations and portfolios that solve each consumer's problem, with $c_t^1 + c_t^2 = y_t^1 + y_t^2$ and $a_{t+1}^1 + a_{t+1}^2 = 0$.

(e) $Q_t = q_{t+1}^0/q_t^0 = \beta$; allocation as in (b). Holdings $a_t^i = \Upsilon_t^i$ (8.8.1):

$$\Upsilon_t^2 = \sum_{\tau \geq t} \beta^{\tau-t} (\mu - y_\tau^2) = \begin{cases} \frac{\mu}{1-\beta} - \frac{\alpha\beta}{1-\beta^2} = 0, & t \text{ even,} \\ \frac{\mu}{1-\beta} - \frac{\alpha}{1-\beta^2} = \frac{\mu}{1-\beta} - \frac{\mu}{\beta(1-\beta)} = -\frac{\mu}{\beta}, & t \text{ odd,} \end{cases} \quad \Upsilon_t^1 = -\Upsilon_t^2.$$

At each even date, type 2 sells μ/β bonds for $\beta \cdot \mu/\beta = \mu$ and consumes it. At each odd date he repays μ/β out of α and consumes $\alpha - \mu/\beta = \mu$. Type 1 lends his whole even-date endowment and consumes $\mu/\beta + \mu = \alpha$ at odd dates. Debt limits: $A_t^2 = \alpha\beta/(1 - \beta^2)$ (even), $\alpha/(1 - \beta^2)$ (odd); $A_t^1 = \mu/(1 - \beta)$. Neither binds.

Numerical exercise — see the companion code

Illustration $\beta = .95$, $\mu = 1$ ($\alpha = 2.0526$): both wealths 20. The notebook checks budgets and KKT conditions on $T = 4000$ periods and the sequential budget along the path.

Exercise 8.10 Corners

Question

pp. 313–314. A pure endowment economy consists of two type of consumers. Consumers of type 1 order consumption streams of the one good according to $\sum_{t=0}^{\infty} \beta^t c_t^1$ and consumers of type 2 order consumption streams according to $\sum_{t=0}^{\infty} \beta^t \ln(c_t^2)$, where $c_t^i \geq 0$ is the consumption of a type i consumer and $\beta \in (0, 1)$ is a common discount factor. Please note the nonnegativity constraint on consumption of each person (the force of this is that c_t^i is *consumption*, not *production*). The consumption good is tradable but nonstorable. There are equal numbers of the two types of consumer. The consumer of type 1 is endowed with the consumption sequence $y_t^1 = \mu > 0 \ \forall t \geq 0$ where $\mu > 0$. The consumer of type 2 is endowed with the consumption sequence $y_t^2 = 0$ if $t \geq 0$ is even, $y_t^2 = \alpha$ if $t \geq 0$ is odd, where

$$\alpha = \mu(1 + \beta^{-1}). \quad (1)$$

- a. Define a competitive equilibrium with time 0 trading. Be careful to include definitions of all of the objects of which a competitive equilibrium is composed.
- b. Compute a competitive equilibrium allocation with time 0 trading. Compute the equilibrium price system. Please also compute the sequence of one-period gross interest rates. Do they differ between odd and even periods?
- c. Compute the time 0 wealths of the two types of consumers using the competitive equilibrium prices.
- d. Now consider an economy identical to the preceding one except in one respect. The endowment of consumer 1 continues to be 1 each period, but we assume that the endowment of consumer 2 is larger (though it continues to be zero in every even period). In particular, we alter the assumption about endowments in condition (1) to the new condition

$$\alpha > \mu(1 + \beta^{-1}).$$

Compute the competitive equilibrium allocation and price system for this economy.

- e. Compute the sequence of one-period interest rates implicit in the equilibrium price system that you computed in part d. Are interest rates higher or lower than those you computed in part b?

Solution:

Misprint. In (d), “continues to be 1 each period” should be “continues to be μ ”.

(a)–(c) As in Exercise 8.9: $q_t^0 = \beta^t$, $c_t^2 = \mu$, $c_t^1 = 0$ (even), α (odd). Gross one-period rates $R_{t,t+1} = q_t^0/q_{t+1}^0 = 1/\beta$ for every t : **no** odd/even difference. Wealths are both $\mu/(1 - \beta)$.

(d) With $q_t^0 = \beta^t$ type 2 would consume $\bar{c} = \alpha\beta/(1 + \beta) > \mu$, which forces $c_t^1 = \mu - \bar{c} < 0$ at even dates. That is infeasible. So at even dates type 1 is at the corner and type 2

consumes everything; at odd dates both are interior. KKT conditions:

$$\text{even } t : c_t^1 = 0, c_t^2 = \mu, \beta^t \leq \mu_1 q_t^0; \quad \text{odd } t : \beta^t = \mu_1 q_t^0, c_t^2 = \frac{\beta^t}{\mu_2 q_t^0} = \frac{\mu_1}{\mu_2} \equiv c_o.$$

Type 2 at even t : $\beta^t/\mu = \mu_2 q_t^0$. With $q_0^0 = 1$, $\mu_2 = 1/\mu$ and so $q_t^0 = \beta^t$ (even). At odd t , $q_t^0 = \beta^t/\mu_1$ and $c_o = \mu_1\mu$. Type 2's budget:

$$\begin{aligned} \mu \sum_{\text{even}} \beta^t + c_o \sum_{\text{odd}} \frac{\beta^t}{\mu_1} &= \alpha \sum_{\text{odd}} \frac{\beta^t}{\mu_1} \\ \iff \frac{\mu}{1 - \beta^2} + \frac{\mu_1 \mu \beta}{\mu_1(1 - \beta^2)} &= \frac{\alpha \beta}{\mu_1(1 - \beta^2)} \iff \mu(1 + \beta) = \frac{\alpha \beta}{\mu_1} \iff \mu_1 = \frac{\alpha \beta}{\mu(1 + \beta)}. \end{aligned}$$

So

$$q_t^0 = \begin{cases} \beta^t & , \\ \beta^t/\mu_1 & , \end{cases} \quad c_t^2 = \begin{cases} \mu & , \\ \frac{\alpha \beta}{1 + \beta} & , \end{cases} \quad c_t^1 = \begin{cases} 0, & t \text{ even}, \\ \mu + \frac{\alpha}{1 + \beta}, & t \text{ odd}, \end{cases} \quad \mu_1 = \frac{\alpha \beta}{\mu(1 + \beta)} > 1.$$

The even-date KKT condition $\beta^t \leq \mu_1 \beta^t$ holds because $\mu_1 > 1 \iff \alpha > \mu(1 + \beta^{-1})$.

✓ Type 1's budget holds by Walras' law.

(e) $R_{t,t+1} = q_t^0/q_{t+1}^0$:

$$\text{even} \rightarrow \text{odd}: \frac{\beta^t}{\beta^{t+1}/\mu_1} = \frac{\mu_1}{\beta} > \frac{1}{\beta}, \quad \text{odd} \rightarrow \text{even}: \frac{\beta^t/\mu_1}{\beta^{t+1}} = \frac{1}{\beta \mu_1} < \frac{1}{\beta}.$$

Rates now alternate: higher than in (b) from even to odd dates, lower from odd to even. The two-period rate is still β^{-2} . Odd-date goods are abundant and cheap. The linear consumer, the only one who could arbitrage intertemporally, is stuck at zero consumption at even dates and cannot lend more, so prices no longer equal β^t .

Numerical exercise — see the companion code

Illustration $\beta = .95$, $\mu = 1$, $\alpha = 1.2\mu(1 + \beta^{-1}) = 2.4632$: $\mu_1 = 1.2$; $c^2 = (1, 1.2)$, $c^1 = (0, 2.2632)$ (even, odd); gross rates 1.26316 (even→odd) and .87719 (odd→even) vs 1.05263 in (b). Wealths: type 1 18.3761, type 2 20. Budgets and KKT conditions are checked on $T = 4000$ periods.

Changes of Measure (8.11–8.12)

Question

pp. 314–315. Let $\{d_t(s_t)\}_{t=0}^\infty$ be a stream of payouts. Suppose that there are complete markets. From (8.5.4) and (8.7.1), the price at time 0 of a claim on this stream of dividends is

$$a_0 = \sum_{t=0} \sum_{s^t} \beta^t \frac{u'(c_t^i(s^t))}{\mu_i} \pi_t(s^t) d_t(s_t).$$

Show that this a_0 can also be represented as

$$a_0 = \sum_t b_t \sum_{s^t} d_t(s_t) \tilde{\pi}_t(s^t) = \tilde{E}_0 \sum_{t=0}^\infty b_t d_t(s_t) \quad (1)$$

where $\tilde{E}$ is the mathematical expectation with respect to the twisted measure $\tilde{\pi}_t(s^t)$ defined by

$$\tilde{\pi}_t(s^t) = b_t^{-1} \beta^t \frac{u'(c_t^i(s^t))}{\mu_i} \pi_t(s^t), \quad b_t = \sum_{s^t} \beta^t \frac{u'(c_t^i(s^t))}{\mu_i} \pi_t(s^t).$$

Prove that $\tilde{\pi}_t(s^t)$ is a probability measure. Interpret b_t itself as a price of particular asset. Note: $\tilde{\pi}_t(s^t)$ is called an *equivalent martingale measure*. See chapters 13 and 14.

Solution:

By (8.5.4), $q_t^0(s^t) = \beta^t \pi_t(s^t) u'(c_t^i(s^t)) / \mu_i$, so $b_t = \sum_{s^t} q_t^0(s^t)$ and $\tilde{\pi}_t(s^t) = q_t^0(s^t) / b_t$.

Probability measure. $q_t^0(s^t) > 0$ whenever $\pi_t(s^t) > 0$ ($u' > 0$, $\mu_i > 0$), so $b_t > 0$ and $\tilde{\pi}_t \geq 0$. Also $\sum_{s^t} \tilde{\pi}_t(s^t) = b_t^{-1} \sum_{s^t} q_t^0(s^t) = 1$. It is *equivalent* to π_t : $\tilde{\pi}_t(s^t) > 0 \iff \pi_t(s^t) > 0$.

Representation. $a_0 = \sum_t \sum_{s^t} q_t^0(s^t) d_t = \sum_t b_t \sum_{s^t} \tilde{\pi}_t(s^t) d_t = \tilde{E}_0 \sum_t b_t d_t$.

Interpretation. $b_t = \sum_{s^t} q_t^0(s^t)$ is the time-0 price of one unit of consumption at t in *every* history: the riskless t -period strip, i.e. a zero-coupon bond (8.7.3). So a_0 is the sum of $\tilde{E}_0 d_t$, the expectation under the risk-neutral measure, discounted at risk-free bond prices. $\tilde{\pi}_t$ tilts probabilities towards histories with high marginal utility.

Numerical exercise — see the companion code

Check in the economy of 8.6 with $d_t =$ endowment, truncated at $T = 10$ (all 2^{10} histories): $\tilde{\pi}_t$ sums to one at each t ; $b_t = e' Q^t \mathbf{1}$; both representations and Exercise 8.12's give $a_0 = 8.839043$.

Exercise 8.12 Harrison–Kreps prices

Question

p. 315. Show that the asset price in (1) of exercise 8.11 can also be represented as

$$a_0 = \sum_{t=0}^{\infty} \sum_{s^t} \beta^t p_t^0(s^t) d_t(s^t) \pi_t(s^t) = E_0 \sum_{t=0}^{\infty} \beta^t p_t^0 d_t$$

where $p_t^0(s^t) = q_t^0(s^t)/[\beta^t \pi_t(s^t)]$.

Solution:

By definition $q_t^0(s^t) = \beta^t p_t^0(s^t) \pi_t(s^t)$. Substitute into (8.7.1):

$$a_0 = \sum_t \sum_{s^t} q_t^0(s^t) d_t(s^t) = \sum_t \sum_{s^t} \beta^t p_t^0(s^t) d_t(s^t) \pi_t(s^t) = E_0 \sum_t \beta^t p_t^0 d_t.$$

By (8.5.4), $p_t^0(s^t) = u'(c_t^i(s^t))/\mu_i$: the Harrison–Kreps state-price density is the marginal utility of consumption in units of time-0 wealth, the same for every consumer. Under the physical measure the payoff is deflated by $\beta^t p_t^0$ (a stochastic discount factor). Under $\tilde{\pi}$ (8.11) it is deflated by the deterministic b_t . The two are linked by $\tilde{\pi}_t = \beta^t p_t^0 \pi_t / b_t$.

Sequential Trading and Market Structures (8.13–8.14)

Exercise 8.13 Early resolution of uncertainty

Question

pp. 315–316. An economy consists of two consumers named $i = 1, 2$. Each consumer evaluates streams of a single consumption good according to $\sum_{t=0}^{\infty} \sum_{s^t} \beta^t u[c_t^i(s^t)] \pi_t(s^t)$. Here $u(c)$ is an increasing, twice continuously differentiable, strictly concave function of consumption c of one good. The utility function satisfies the Inada condition $\lim_{c \downarrow 0} u'(c) = +\infty$. A feasible allocation satisfies $\sum_i c_t^i(s^t) \leq \sum_i y^i(s_t)$. The consumers' endowments of the one nonstorable good are both functions of a state variable $s_t \in \mathbf{S} = \{0, 1, 2\}$; s_t is described by a time invariant Markov chain with initial distribution $\pi_0 = [0 \ 1 \ 0]'$ and transition density defined by the stochastic matrix

$$P = \begin{bmatrix} 1 & 0 & 0 \\ .5 & 0 & .5 \\ 0 & 0 & 1 \end{bmatrix}.$$

The endowments of the two consumers are

$$y_t^1 = s_t/2, \quad y_t^2 = 1 - s_t/2.$$

- Define a competitive equilibrium with Arrow securities.
- Compute a competitive equilibrium with Arrow securities.
- By hand, simulate the economy. In particular, for every possible realization of the histories s^t , describe time series of c_t^1, c_t^2 and the wealth levels a_t^i of the consumers. (Note: Usually this would be an impossible task by hand, but this problem has been set up to make the task manageable.)

Solution:

$s_0 = 1$ for sure. At $t = 1$ the chain moves to 0 or 2 (probability .5 each) and stays there forever. So there are two histories, $(1, 0, 0, \dots)$ and $(1, 2, 2, \dots)$. The aggregate endowment is $s/2 + 1 - s/2 = 1$ in every state.

(a) A recursive equilibrium with Arrow securities (section 8.9.3): a kernel $Q(s'|s)$; natural debt limits $\bar{A}^i(s)$; value functions $v^i(a, s)$; decision rules $h^i(a, s)$, $g^i(a, s, s')$; initial wealths $a_0^i = 0$. Given Q and $\bar{A}^i$, the rules solve $v^i(a, s) = \max\{u(c) + \beta \sum_{s'} v^i(\hat{a}(s'), s') P(s'|s)\}$ s.t. $c + \sum_{s'} Q(s'|s) \hat{a}(s') \leq y^i(s) + a$, $-\hat{a}(s') \leq \bar{A}^i(s')$; $\bar{A}^i = y^i + Q\bar{A}^i$; and markets clear: $c^1 + c^2 = 1$, $\hat{a}^1(s') + \hat{a}^2(s') = 0$.

(b) With a constant aggregate endowment, $u'(c_t^1)/u'(c_t^2)$ constant (8.5.5) and $c^1 + c^2 = 1$ give constant consumptions, so (8.9.12) gives $Q(s'|s) = \beta P(s'|s)$. Consumer 1's time-0 wealth is

$$y^1(1) + \sum_{t \geq 1} \beta^t [.5 y^1(0) + .5 y^1(2)] = .5 + \frac{.5\beta}{1-\beta} = \frac{.5}{1-\beta},$$

and by symmetry consumer 2's is the same. Hence $c_t^1 = c_t^2 = .5$ always. Holdings

$\hat{a}^i(s) = \Upsilon^i(s)$:

$$\Upsilon^1(0) = \frac{.5 - 0}{1 - \beta}, \quad \Upsilon^1(2) = \frac{.5 - 1}{1 - \beta}, \quad \Upsilon^1(1) = 0, \quad \Upsilon^2 = -\Upsilon^1.$$

Debt limits: $\bar{A}^1 = (I - \beta P)^{-1} y^1 = (0, \frac{.5}{1-\beta}, \frac{1}{1-\beta})$ and $\bar{A}^2 = (\frac{1}{1-\beta}, \frac{.5}{1-\beta}, 0)$ over $s = (0, 1, 2)$. Neither binds: the consumer with a zero limit in a state holds positive claims there.

(c) At $t = 0$ ($s = 1$), consumer 1 buys $.5/(1 - \beta)$ claims on $s' = 0$ and sells $.5/(1 - \beta)$ claims on $s' = 2$. The cost is $.5\beta \cdot .5/(1 - \beta) - .5\beta \cdot .5/(1 - \beta) = 0$, and he consumes his endowment $.5$. Consumer 2 takes the opposite positions.

history	c_t^1, c_t^2	$a_t^1 (t \geq 1)$	$a_t^2 (t \geq 1)$	endowments ($t \geq 1$)
1, 0, 0, ...	.5, .5	$.5/(1 - \beta)$	$-.5/(1 - \beta)$	$y^1 = 0, y^2 = 1$
1, 2, 2, ...	.5, .5	$-.5/(1 - \beta)$	$.5/(1 - \beta)$	$y^1 = 1, y^2 = 0$

After $t = 1$ each consumer rolls his position over: in history $(1, 0, \dots)$ consumer 1 receives a , consumes $.5$, and reinvests $\beta a = a - .5$. At $t = 0$ all $a_0^i = 0$. All risk is resolved at $t = 1$ and was fully insured at $t = 0$.

Numerical exercise — see the companion code

Illustration $\beta = .95$: $\Upsilon^1 = (10, 0, -10)$, $\bar{A}^1 = (0, 10, 20)$. The Arrow budget holds in all three states.

Exercise 8.14 Donated by Pierre-Olivier Weill

Question

pp. 316–317. An economy is populated by a continuum of infinitely lived consumers of types $j \in \{0, 1\}$, with a measure one of each. There is one nonstorable consumption good arriving in the form of an endowment stream owned by each consumer. Specifically, the endowments are

$$y_t^0(s_t) = (1 - s_t)\bar{y}^0, \quad y_t^1(s_t) = s_t\bar{y}^1,$$

where s_t is a two-state time-invariant Markov chain valued in $\{0, 1\}$ and $\bar{y}^0 < \bar{y}^1$. The initial state is $s_0 = 1$. Transition probabilities are denoted by $\pi(s'|s)$ for $(s, s') \in \{0, 1\}^2$, where $'$ denotes a next period value. The aggregate endowment is $y_t(s_t) \equiv (1 - s_t)\bar{y}^0 + s_t\bar{y}^1$. Thus, this economy fluctuates stochastically between recessions $y_t(0) = \bar{y}^0$ and booms $y_t(1) = \bar{y}^1$. In a recession, the aggregate endowment is owned by type 0 consumers, while in a boom it is owned by a type 1 consumers. A consumer orders consumption streams according to:

$$U(c^j) = \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi(s^t|s_0) \frac{c_t^j(s^t)^{1-\gamma}}{1-\gamma},$$

where $s^t = (s_t, s_{t-1}, \dots, s_0)$ is the history of the state up to time t , $\beta \in (0, 1)$ is the discount factor, and $\gamma > 0$ is the coefficient of relative risk aversion.

- a. Define a competitive equilibrium with time 0 trading. Compute the price system $\{q_t^0(s^t)\}_{t=0}^{\infty}$ and the equilibrium allocation $\{c^j(s^t)\}_{t=0}^{\infty}$, for $j \in \{0, 1\}$.
- b. Find a utility function $\bar{U}(c) = E_0\left(\sum_{t=0}^{\infty} \beta^t u(c_t)\right)$ such that the price system $q_t^0(s^t)$ and the aggregate endowment $y_t(s_t)$ is an equilibrium allocation of the single-agent economy $(\bar{U}, \{y_t(s_t)\}_{t=0}^{\infty})$. How does your answer depend on the initial distribution of endowments $y_t^j(s_t)$ among the two types $j \in \{0, 1\}$? How would you defend the representative agent assumption in this economy?
- c. Describe the equilibrium allocation under the following three market structures:
 - (i) at each node s^t , agents can trade only claims on their entire endowment streams;
 - (ii) at each node s^t , there is a complete set of one-period ahead Arrow securities;
 - and (iii) at each node s^t , agents can only trade two risk-free assets, namely, a one-period zero-coupon bond that pays one unit of consumption for sure at $t + 1$ and a two-period zero-coupon bond that pays one unit of the consumption good for sure at $t + 2$. How would you modify your answer in the absence of aggregate uncertainty?
- d. Assume that $\pi(1|0) = 1$, $\pi(0|1) = 1$, and as before $s_0 = 1$. Compute the allocation in an equilibrium with time 0 trading. Does the type $j = 1$ agent always consume the largest share of the aggregate endowment? How does it depend on parameter values? Provide economic intuition for your results.
- e. Assume that $\pi(1|0) = 1$ and $\pi(0|1) = 1$. Remember that $s_0 = 1$. Assume that at $t = 1$ agent $j = 0$ is given the option to default on her financial obligation. For example, in the time 0 trading economy, these obligations are deliveries of goods. Upon default, it is assumed that the agent is excluded from the market and has to consume her endowment forever. Will the agent ever exercise her option to default?

Solution:

Misprint. “booms $y_t(1) = \bar{y}^0$ ” should be $y_t(1) = \bar{y}^1$.

(a) Definition: a price system $\{q_t^0(s^t)\}$ and allocations $\{c^0, c^1\}$ such that each type maximises $U(c^j)$ s.t. $\sum_t \sum_{s^t} q_t^0 c_t^j \leq \sum_t \sum_{s^t} q_t^0 y_t^j$, and $c_t^0 + c_t^1 = y_t(s^t)$. With identical CRRA utility, (8.6.1) gives constant shares, $c_t^j(s^t) = \theta_j y_t(s^t)$, and (*) gives

$$q_t^0(s^t) = \beta^t \pi(s^t | s_0) \left(\frac{y_t(s^t)}{\bar{y}^1} \right)^{-\gamma} \quad (y_0 = \bar{y}^1).$$

Budgets: the value of c^j is $\theta_j W$, where $W = \sum_t \sum_{s^t} q_t^0 y_t$. Hence $\theta_j = W^j / W$ with $W^j = \sum_t \sum_{s^t} q_t^0 y_t^j$. In matrix form (states ordered 0, 1), with $m(s) = (y(s)/\bar{y}^1)^{-\gamma}$,

$$\theta_0 = \frac{e'_1 (I - \beta P)^{-1} (m \odot y^0)}{e'_1 (I - \beta P)^{-1} (m \odot y)}, \quad y^0 = (\bar{y}^0, 0)', \quad y = (\bar{y}^0, \bar{y}^1)', \quad \theta_1 = 1 - \theta_0.$$

(b) $\bar{U}(c) = E_0 \sum_t \beta^t c_t^{1-\gamma} / (1 - \gamma)$, the common utility. The single agent consumes y_t , and his FOC gives exactly $q_t^0 = \beta^t \pi_t(y_t/y_0)^{-\gamma}$. The answer does *not* depend on the distribution of endowments: redistribution changes θ_j but not prices, because $u'(\theta_j y) = \theta_j^{-\gamma} u'(y)$ scales all marginal utilities by the same factor (Gorman aggregation; compare v_θ in Exercise 8.3, here $v_\theta(c) \propto c^{1-\gamma} / (1 - \gamma)$). Defence: complete markets plus identical homothetic preferences, so heterogeneity in endowments washes out of prices.

(c) (i) *Claims on endowment streams.* The equilibrium allocation $c^j = \theta_j y$ is the payoff of a portfolio holding fraction θ_j of *both* endowment claims. At $t = 0$, type 0 sells θ_1 of his claim and buys θ_0 of type 1's. The cost is $\theta_0(W^0 + W^1) - W^0 = 0$, so the trade is affordable. After that no trade is needed. Same allocation. (ii) *Arrow securities:* same allocation (section 8.8.6). (iii) *Two risk-free bonds.* At $t + 1$ the two-period bond is a one-period bond worth $p_1(s_{t+1}) = \beta \sum_{s''} \pi(s'' | s_{t+1}) (y(s'')/y(s_{t+1}))^{-\gamma}$. The payoff matrix at $t + 1$ across $s_{t+1} \in \{0, 1\}$ is $\begin{bmatrix} 1 & p_1(0) \\ 1 & p_1(1) \end{bmatrix}$. It is nonsingular iff $p_1(0) \neq p_1(1)$,

and then the two bonds span both Arrow securities. Markets are dynamically complete and the allocation is the same. *No aggregate uncertainty* (y constant): (i) and (ii) still deliver $c^j = \theta_j y$, constant. In (iii), $p_1(s) = \beta$ in both states, so the matrix has rank one: only risk-free borrowing and lending is possible. The idiosyncratic endowment risk cannot be insured, and the complete-markets allocation is generally *not* attained (an incomplete-markets economy, chapters 17–18).

(d) Deterministic alternation $s = 1, 0, 1, 0, \dots$: $y_t = \bar{y}^1$ (even t), $\bar{y}^0$ (odd t), and $q_t^0 = \beta^t (y_t/\bar{y}^1)^{-\gamma}$. Wealths:

$$W^1 = \bar{y}^1 \sum_{k \geq 0} \beta^{2k} = \frac{\bar{y}^1}{1 - \beta^2}, \quad W^0 = \sum_{k \geq 0} \beta^{2k+1} \left(\frac{\bar{y}^0}{\bar{y}^1} \right)^{-\gamma} \bar{y}^0 = \frac{\beta \bar{y}^1 (\bar{y}^0/\bar{y}^1)^{1-\gamma}}{1 - \beta^2},$$

$$\theta_1 = \frac{1}{1 + \beta (\bar{y}^0/\bar{y}^1)^{1-\gamma}}, \quad c_t^1 = \theta_1 y_t, \quad c_t^0 = (1 - \theta_1) y_t.$$

Type 1 gets the larger share iff $\beta(\bar{y}^0/\bar{y}^1)^{1-\gamma} < 1$. If $\gamma \leq 1$ this always holds, since $\bar{y}^0/\bar{y}^1 < 1$ and $\beta < 1$. If $\gamma > 1$ it holds iff

$$\gamma < 1 + \frac{\ln(1/\beta)}{\ln(\bar{y}^1/\bar{y}^0)}.$$

Intuition: two forces. Type 1's endowment comes first and is larger (favours type 1). But it arrives in booms, when goods are cheap. With high curvature the price of recession goods, $\propto \bar{y}^{0-\gamma}$, is so high that type 0's small recession endowment is worth more.

(e) At $t = 1$ (recession) type 0 must deliver $\theta_1 \bar{y}^0$ of her $\bar{y}^0$. Staying gives

$$V^{\text{stay}} = \sum_{k \geq 0} \beta^{2k} [u(\theta_0 \bar{y}^0) + \beta u(\theta_0 \bar{y}^1)] = \frac{\theta_0^{1-\gamma} [(\bar{y}^0)^{1-\gamma} + \beta (\bar{y}^1)^{1-\gamma}]}{(1-\gamma)(1-\beta^2)}.$$

Defaulting gives autarky, $\bar{y}^0$ at odd dates and 0 at even dates: $V^{\text{def}} = \sum_k \beta^{2k} [u(\bar{y}^0) + \beta u(0)]$. If $\gamma \geq 1$, $u(0) = -\infty$, so she **never** defaults. If $\gamma < 1$, $u(0) = 0$. Let $r = (\bar{y}^0/\bar{y}^1)^{1-\gamma}$ and $\theta_0 = \beta r/(1+\beta r)$ from (d). Multiplying by $(1-\gamma)(1-\beta^2)/(\bar{y}^1)^{1-\gamma} > 0$, she defaults iff

$$r > \theta_0^{1-\gamma}(r + \beta).$$

At $\gamma \rightarrow 0$ this becomes $1 + \beta x > \beta x + \beta^2$ (with $x = \bar{y}^0/\bar{y}^1$), which always holds: a risk-neutral agent defaults. At $\gamma \rightarrow 1$ it becomes $1 > 1 + \beta$, which never holds. So default occurs only for curvature below a threshold $\gamma^*(x) \in (0, 1)$ (figure). Autarky means consumption alternating between $\bar{y}^0$ and 0, a disaster unless the agent barely cares about smoothing.

Numerical exercise — see the companion code

Illustration $\bar{y}^0 = 1$, $\bar{y}^1 = 2$, $P = \begin{bmatrix} .7 & .3 \\ .2 & .8 \end{bmatrix}$ (rows $s = 0, 1$), $\beta = .95$, $\gamma = 2$: $\theta_0 = .531469$ (matrix inverse = truncated sum). Bond prices $p_1 = (.73625, 1.52)$, so the (iii) payoff matrix has rank 2. (d) with $\beta = .95$: $\theta_1 = .5982, .3448, .0318$ for $\gamma = .5, 2, 6$; $\theta_1 < \frac{1}{2}$ iff $\gamma > 1.0740$. (e): the closed form equals truncated utility sums. Thresholds $\gamma^* = .048, .061, .068$ at $x = .2, .5, .8$.

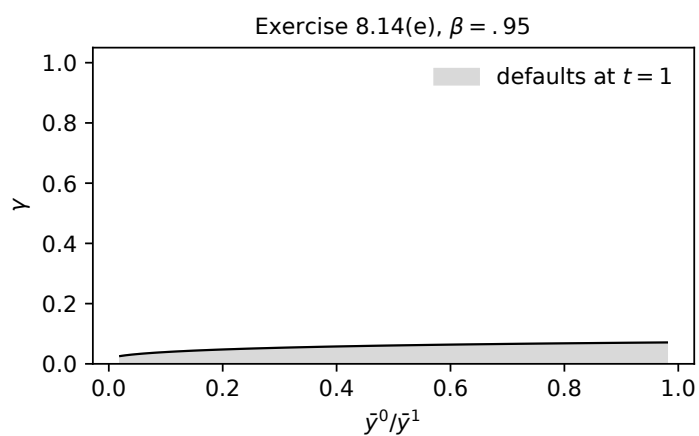


Figure 1: Exercise 8.14(e): parameter region in which type 0 defaults at $t = 1$ ($\beta = .95$).

Heterogeneous Beliefs (8.15–8.19)

Exercise 8.15 Diverse beliefs, I

Question

pp. 317–318. A pure endowment economy is populated by two consumers. Consumer i has preferences over history-contingent consumption sequences $\{c_t^i(s^t)\}$ that are ordered by

$$\sum_{t=0}^{\infty} \sum_{s^t} \beta^t u(c_t^i(s^t)) \pi_t^i(s^t),$$

where $u(c) = \ln(c)$ and where $\pi_t^i(s^t)$ is a density that consumer i assigns to history s^t . The state space is time invariant. In particular, $s_t \in S = \{0, .5, 1\}$ for all $t \geq 0$. Only two histories are possible for $t = 0, 1, 2, \dots$:

history 1 : .5, 1, 1, 1, 1, ... history 2 : .5, 0, 0, 0, 0, ...

Consumer 1 assigns probability 1/3 to history 1 and probability 2/3 to history 2, while consumer 2 assigns probability 2/3 to history 1 and probability 1/3 to history 2. Nature assigns equal probabilities to the two histories. The endowments of the two consumers are:

$$y_t^1 = s_t, \quad y_t^2 = 1 - s_t.$$

- Define a competitive equilibrium with sequential trading of a complete set of one-period Arrow securities.
- Compute a competitive equilibrium with sequential trading of a complete set of one-period Arrow securities.
- Is the equilibrium allocation Pareto optimal?

Solution:

(a) As in section 8.8.5, with subjective probabilities. Objects: pricing kernels $\tilde{Q}_t(s_{t+1}|s^t)$; natural debt limits $A_t^i(s^t)$ (8.8.2); an allocation $\{c_t^i(s^t)\}$ and portfolios $\{\tilde{a}_{t+1}^i(s_{t+1}, s^t)\}$; initial wealth $\tilde{a}_0^i = 0$. Given kernels and limits, each consumer maximises his subjective expected utility subject to (8.8.3)–(8.8.4). Markets clear: $c^1 + c^2 = 1$, $\tilde{a}^1 + \tilde{a}^2 = 0$. Interior optimality gives $\tilde{Q}_t(s_{t+1}|s^t) = \beta \pi_t^i(s^{t+1}|s^t) c_t^i(s^t) / c_{t+1}^i(s^{t+1})$ for $i = 1, 2$.

(b) The aggregate endowment is 1 always. After $t = 1$ there is no uncertainty, so within each history consumption is constant and $\tilde{Q} = \beta$. At $t = 0$ the Euler equations for $s_1 = 1$ (history 1) and $s_1 = 0$ (history 2) are

$$\tilde{Q}(1) = \beta \frac{1}{3} \frac{c_0^1}{c^1(h_1)} = \beta \frac{2}{3} \frac{c_0^2}{c^2(h_1)}, \quad \tilde{Q}(0) = \beta \frac{2}{3} \frac{c_0^1}{c^1(h_2)} = \beta \frac{1}{3} \frac{c_0^2}{c^2(h_2)}.$$

Let $k = c_0^1/c_0^2$. Then $c^1(h_1)/c^2(h_1) = k/2$ and $c^1(h_2)/c^2(h_2) = 2k$, so $c_0^1 = \frac{k}{1+k}$, $c^1(h_1) = \frac{k}{2+k}$, $c^1(h_2) = \frac{2k}{1+2k}$. Equivalently, with time-0 prices $q_t(h) = \beta^t \pi^1(h) / (\mu_1 c^1(h))$ and $\mu_1 = 1/c_0^1 = (1+k)/k$, consumer 1's budget (log utility: the value of consumption is

$1/[\mu_1(1 - \beta)]$ reads

$$\begin{aligned} \frac{1}{\mu_1(1 - \beta)} &= .5 + \frac{\beta}{1 - \beta} \frac{1/3}{\mu_1 c^1(h_1)} \\ \iff \frac{1}{1 - \beta} &= \frac{1 + k}{2k} + \frac{\beta(2 + k)}{3k(1 - \beta)} \\ \iff 6k &= 3(1 + k)(1 - \beta) + 2\beta(2 + k) \iff k(3 + \beta) = 3 + \beta. \end{aligned}$$

So $k = 1$ for every β , and

$$\boxed{\begin{aligned} c_0^1 = c_0^2 = .5; \quad c^1(h_1) = \frac{1}{3}, \quad c^2(h_1) = \frac{2}{3}; \quad c^1(h_2) = \frac{2}{3}, \quad c^2(h_2) = \frac{1}{3}; \\ \tilde{Q}_0(1) = \tilde{Q}_0(0) = \frac{\beta}{2}, \quad \tilde{Q}_t = \beta \quad (t \geq 1). \end{aligned}}$$

Holdings $\tilde{a} = \Upsilon$: $a^1(h_1) = (\frac{1}{3} - 1)/(1 - \beta) = -\frac{2/3}{1 - \beta}$ and $a^1(h_2) = \frac{2/3}{1 - \beta}$, with $a^2 = -a^1$. The $t=0$ budget: $.5 + \frac{\beta}{2}(a^1(h_1) + a^1(h_2)) = .5$. ✓ Each consumer *bets on his beliefs*. Consumer 1 thinks history 2 is likely, so he buys claims on it and ends up consuming $2/3$ there, although his endowment there is zero. The natural debt limits (zero for consumer 1 in h_2 and for consumer 2 in h_1) do not bind.

(c) Yes, relative to the consumers' own preferences. The allocation satisfies (8.B.2) with $\lambda_1 = \lambda_2$: on h_1 , $\frac{\pi^2}{\pi^1} \frac{u'(c^2)}{u'(c^1)} = 2 \cdot \frac{1/3}{2/3} = 1$, and likewise on h_2 . So it solves the Pareto problem with subjective expected utilities (first welfare theorem). Nature's probabilities play no role in Pareto optimality. Measured with the true $(\frac{1}{2}, \frac{1}{2})$, each consumer's consumption is risky: that is the price of disagreement.

Numerical exercise — see the companion code

brentq on consumer 1's budget gives $k = 1$ for $\beta \in \{.5, .9, .95\}$. With $\beta = .95$: both consumers' Euler equations give $\tilde{Q}_0 = (.475, .475)$ and $a^1 = (-13.333, 13.333)$.

Exercise 8.16 Diverse beliefs, II

Question

pp. 318–319. Consider the following I person pure endowment economy. There is a state variable $s_t \in S$ for all $t \geq 0$. Let s^t denote a history of s from 0 to t . The time t aggregate endowment is a function of the history, so $Y_t = Y_t(s^t)$. Agent i attaches a personal probability of $\pi_t^i(s^t)$ to history s^t . The history s^t is observed by all I people at time t . Assume that for all i , $\pi_t^i(s_t) > 0$ if and only if $\pi_t^1(s_t) > 0$ (so the consumers agree about which histories have positive probability). Consumer i ranks consumption plans $c_t^i(s^t)$ that are measurable functions of histories via the expected utility functional

$$\sum_{t=0}^{\infty} \sum_{s^t} \beta^t \ln(c_t^i(s^t)) \pi_t^i(s^t) \quad (1)$$

The ownership structure of the economy is not yet determined.

A planner puts positive Pareto weights $\lambda_i > 0$ on consumers $i = 1, \dots, I$ and solves a time 0 Pareto problem that respects each consumer's preferences as represented by (1).

- a. Show how to solve for a Pareto optimal allocation. Display an expression for $c_t^i(s^t)$ as a function of $Y_t(s^t)$ and other pertinent variables.
- b. Under what circumstances does the Pareto plan imply complete risk-sharing among the I consumers?
- c. Under what circumstances does the Pareto plan imply an allocation that is not history dependent? By 'not history dependent', we mean that $Y_t(s^t) = Y_t(s^t)$ would imply the same allocation at time t ?
- d. For a given set of Pareto weights, find an associated equilibrium price vector and an initial distribution of wealth among the I consumers that makes the Pareto allocation be the allocation associated with a competitive equilibrium with time 0 trading of history-contingent claims on consumption.
- e. Find a formula for the equilibrium price vector in terms of equilibrium quantities and the beliefs of consumers.
- f. Suppose that $I = 2$. Show that as $\lambda_2/\lambda_1 \rightarrow +\infty$, the planner would distribute initial wealth in a way that makes consumer 2's beliefs more and more influential in determining equilibrium prices.

Solution:

(a) Lagrangian: $\mathcal{L} = \sum_t \sum_{s^t} \left\{ \sum_i \lambda_i \beta^t \ln c_t^i(s^t) \pi_t^i(s^t) + \theta_t(s^t) [Y_t(s^t) - \sum_i c_t^i(s^t)] \right\}$. FOC: $\lambda_i \beta^t \pi_t^i(s^t) / c_t^i(s^t) = \theta_t(s^t)$, so $c_t^i = \lambda_i \beta^t \pi_t^i / \theta_t$. Summing over i and using feasibility, $\theta_t(s^t) = \beta^t \sum_j \lambda_j \pi_t^j(s^t) / Y_t(s^t)$. Hence

$$c_t^i(s^t) = \frac{\lambda_i \pi_t^i(s^t)}{\sum_{j=1}^I \lambda_j \pi_t^j(s^t)} Y_t(s^t).$$

Shares are weighted by Pareto weight times the consumer's probability of the history

(likelihood ratios, as in (8.B.2)).

(b) Complete risk sharing means that each consumer's share c_t^i/Y_t is the same across all histories. The share is $\lambda_i L_t^i / \sum_j \lambda_j L_t^j$ with $L_t^j = \pi_t^j / \pi_t^1$. It is constant across s^t iff each $L_t^j(s^t)$ is constant across s^t . Since $\sum_{s^t} \pi_t^j = \sum_{s^t} \pi_t^1 = 1$, the constant must be 1. So there is complete risk sharing iff **beliefs are common**, $\pi_t^i = \pi_t^1$ for all i (on histories with positive probability). The shares are then $\lambda_i / \sum_j \lambda_j$.

(c) $c_t^i(s^t)$ depends on s^t only through $Y_t(s^t)$ iff every likelihood ratio $\pi_t^j(s^t)/\pi_t^1(s^t)$ is a function of $(t, Y_t(s^t))$ only. Sufficiency is clear from (a). For necessity, $c_t^j/c_t^1 = \lambda_j L_t^j / \lambda_1$ must then be a function of Y_t . Common beliefs are the leading case. Disagreement is harmless only about events that the aggregate endowment reveals.

(d) Prices = normalised multipliers (section 8.5.2), with $q_0^0(s_0) = 1$ and $\pi_0^j(s_0) = 1$:

$$q_t^0(s^t) = \frac{\theta_t(s^t)}{\theta_0(s_0)} = \beta^t \frac{Y_0}{Y_t(s^t)} \sum_j \omega_j \pi_t^j(s^t), \quad \omega_j \equiv \frac{\lambda_j}{\sum_k \lambda_k}.$$

Consumer i 's consumption is worth $\sum_t \sum_{s^t} q_t^0 c_t^i = \sum_t \beta^t Y_0 \omega_i \sum_{s^t} \pi_t^i = \omega_i Y_0 / (1 - \beta)$, and total wealth is $\sum_t \sum_{s^t} q_t^0 Y_t = Y_0 / (1 - \beta)$. An initial distribution that supports the plan gives consumer i wealth $\omega_i Y_0 / (1 - \beta)$. For example, endow consumer i with the fraction $\omega_i = \lambda_i / \sum_k \lambda_k$ of the aggregate endowment in every history. Each consumer's FOC $\beta^t \pi_t^i / c_t^i = \mu_i q_t^0$ holds with $\mu_i = 1/c_0^i$.

(e) From $\beta^t \pi_t^i / c_t^i = q_t^0 / c_0^i$, we get $q_t^0 c_t^i = \beta^t c_0^i \pi_t^i$. Summing over i :

$$q_t^0(s^t) = \beta^t \frac{1}{Y_t(s^t)} \sum_i c_0^i \pi_t^i(s^t) = \beta^t \frac{Y_0}{Y_t(s^t)} \sum_i \frac{c_0^i}{Y_0} \pi_t^i(s^t).$$

This is the marginal utility of the aggregate endowment times a consumption-share-weighted average of beliefs.

(f) With $I = 2$, $\omega_2 = \frac{\lambda_2/\lambda_1}{1+\lambda_2/\lambda_1} \rightarrow 1$, so consumer 2's initial wealth share $\rightarrow 1$ and $q_t^0 \rightarrow \beta^t (Y_0/Y_t) \pi_t^2(s^t)$: prices are those of an economy in which everyone holds consumer 2's beliefs.

Numerical exercise — see the companion code

Random example ($I = 3$, four date-1 nodes, $\lambda = (1, 2, .5)$, $\beta = .9$): SLSQP on the planner's node-by-node problem reproduces (a); every consumer's FOC holds at the prices of (e).

Exercise 8.17 Diverse beliefs, III

Question

pp. 319–320. An economy consists of two consumers named $i = 1, 2$. Each consumer evaluates streams of a single nonstorable consumption good according to

$$\sum_{t=0}^{\infty} \sum_{s^t} \beta^t \ln[c_t^i(s^t)] \pi_t^i(s^t).$$

Here $\pi_t^i(s^t)$ is consumer i 's subjective probability over history s^t . A feasible allocation satisfies $\sum_i c_t^i(s^t) \leq \sum_i y^i(s_t)$ for all $t \geq 0$ and for all s^t . The consumers' endowments of the one good are functions of a state variable $s_t \in \mathbf{S} = \{0, 1, 2\}$. In truth, s_t is described by a time invariant Markov chain with initial distribution $\pi_0 = [0 \ 1 \ 0]'$ and transition density defined by the stochastic matrix

$$P = \begin{bmatrix} 1 & 0 & 0 \\ .5 & 0 & .5 \\ 0 & 0 & 1 \end{bmatrix}$$

where $P_{ij} = \text{Prob}[s_{t+1} = j - 1 | s_t = i - 1]$ for $i = 1, 2, 3$ and $j = 1, 2, 3$. The endowments of the two consumers are

$$y_t^1 = s_t/2, \quad y_t^2 = 1 - s_t/2.$$

In part I, both consumers know the true probabilities over histories s^t (i.e., they know both π_0 and P). In part II, the two consumers have different subjective probabilities.

Part I: Assume that both consumers know (π_0, P) , so that $\pi_t^1(s^t) = \pi_t^2(s^t)$ for all $t \geq 0$ for all s^t .

- Show how to deduce $\pi_t^i(s^t)$ from (π_0, P) .
- Define a competitive equilibrium with sequential trading of Arrow securities.
- Compute a competitive equilibrium with sequential trading of Arrow securities.
- By hand, simulate the economy. In particular, for every possible realization of the histories s^t , describe time series of c_t^1, c_t^2 and the wealth levels for the two consumers.

Part II: Now assume that while consumer 1 knows (π_0, P) , consumer 2 knows π_0 but thinks that P is

$$\hat{P} = \begin{bmatrix} 1 & 0 & 0 \\ .4 & 0 & .6 \\ 0 & 0 & 1 \end{bmatrix}.$$

- Deduce $\pi_t^2(s^t)$ from $(\pi_0, \hat{P})$ for all $t \geq 0$ for all s^t .
- Formulate and solve a Pareto problem for this economy.
- Define an equilibrium with time 0 trading of a complete set of Arrow-Debreu history-contingent securities.
- Compute an equilibrium with time 0 trading of a complete set of Arrow-Debreu history-contingent securities.
- Compute an equilibrium with sequential trading of Arrow securities. For every possible realization of s^t for all $t \geq 0$, please describe time series of c_t^1, c_t^2 and the wealth levels for the two consumers.

Solution:

Part I. (a) $\pi_t(s^t) = \pi_0(s_0) \prod_{k=1}^t P(s_k | s_{k-1})$. Here $s_0 = 1$ surely. For $t \geq 1$, $\pi_t(1, 0, \dots, 0) = \pi_t(1, 2, \dots, 2) = .5$, and every other history has probability zero. **(b)–(d)** This is Exercise 8.13 with $u = \ln$. The definition is 8.13(a) with $u = \ln$. Solution: $Q(s'|s) = \beta P(s'|s)$, $c^1 = c^2 = .5$ in every history. Holdings: consumer 1 has $a^1 = +.5/(1 - \beta)$ on $(1, 0, 0, \dots)$ and $-.5/(1 - \beta)$ on $(1, 2, 2, \dots)$ for $t \geq 1$, and zero at $t = 0$; $a^2 = -a^1$. The table of 8.13(c) applies unchanged.

Part II. (e) $\pi_0^2(1) = 1$; for $t \geq 1$, $\pi_t^2(1, 0, \dots, 0) = .4$ and $\pi_t^2(1, 2, \dots, 2) = .6$; zero otherwise.

(f) $\max \lambda U_1 + (1 - \lambda) U_2$ s.t. $c^1 + c^2 \leq 1$. By 8.16(a), $c_t^1 = \lambda \pi_t^1 / (\lambda \pi_t^1 + (1 - \lambda) \pi_t^2)$:

$$c_0^1 = \lambda, \quad c^1(\text{path } 0) = \frac{.5\lambda}{.5\lambda + .4(1 - \lambda)} = \frac{.5\lambda}{.4 + .1\lambda}, \quad c^1(\text{path } 2) = \frac{.5\lambda}{.6 - .1\lambda},$$

with $c^2 = 1 - c^1$, constant along each path for $t \geq 1$.

(g) Prices $\{q_t^0(s^t)\}$ and an allocation such that each consumer maximises his *subjective* expected utility subject to $\sum_t \sum_{s^t} q_t^0(c_t^i - y_t^i) \leq 0$, and $c^1 + c^2 = 1$.

(h) By 8.16(d) with $Y \equiv 1$ and weights $(\lambda, 1 - \lambda)$: $q_t^0(\text{path } 0) = \beta^t(.4 + .1\lambda)$ and $q_t^0(\text{path } 2) = \beta^t(.6 - .1\lambda)$ for $t \geq 1$. Consumer 1's wealth is $\lambda/(1 - \beta)$ (8.16(d)). His endowment is .5 at $t = 0$, 0 on path 0, and 1 on path 2:

$$\frac{\lambda}{1 - \beta} = .5 + \frac{\beta}{1 - \beta}(.6 - .1\lambda) \iff \lambda(1 + .1\beta) = .5 + .1\beta \iff \boxed{\lambda = \frac{.5 + .1\beta}{1 + .1\beta} > \frac{1}{2}}.$$

Consumer 2 is optimistic about path 2, where consumer 1 owns everything. That bids up the value of consumer 1's endowment and makes him richer. Yet consumer 1 consumes more on path 0, which he thinks more likely than consumer 2 does.

(i) Kernel at $s = 1$: $Q(0|1) = \beta(.4 + .1\lambda)$, $Q(2|1) = \beta(.6 - .1\lambda)$; $Q(s|s) = \beta$ for $s \in \{0, 2\}$. Consumption as in (f). Wealth: $a_0^i = 0$. For $t \geq 1$, $a^1 = c^1(\text{path } 0)/(1 - \beta)$ on path 0 and $a^1 = [c^1(\text{path } 2) - 1]/(1 - \beta)$ on path 2; $a^2 = -a^1$. At $t = 0$ consumer 1 consumes $\lambda > .5$, selling claims on path 2 and buying claims on path 0.

Numerical exercise — see the companion code

$\beta = .95$ (not given): $\lambda = .543379$; $Q(0|1) = .431621$, $Q(2|1) = .518379$; $c^1 = (.59799, .49791)$ on paths $(0, 2)$; $a^1 = (11.9598, -10.0418)$. `brentq` on the budget recovers λ . Both Euler equations and both $t = 0$ Arrow budgets hold.

Exercise 8.18 Diverse beliefs, IV

Question

pp. 320–321. A pure exchange economy is populated by two consumers. Consumer i has preferences over history-contingent consumption sequences $\{c_t^i(s^t)\}$ that are ordered by

$$\sum_{t=0}^{\infty} \sum_{s^t} \beta^t u(c_t^i(s^t)) \pi_t^i(s^t),$$

where $u(c) = \ln(c)$, $\beta \in (0, 1)$, and $\pi_t^i(s^t)$ is a density that consumer i assigns to history s^t . The state space is time invariant. In particular, $s_t \in S = \{0, .5, 1\}$ for all $t \geq 0$. Only two histories are possible for $t = 0, 1, 2, \dots$:

history 1 : .5, 1, 1, 1, 1, ... history 2 : .5, 0, 0, 0, 0, ...

Consumer 1 assigns probability 1 to history 1 and probability 0 to history 2, while consumer 2 assigns probability 0 to history 1 and probability 1 to history 2. Nature assigns equal probabilities to the two histories. The endowments of the two consumers are:

$$y_t^1 = s_t, \quad y_t^2 = 1 - s_t.$$

- Formulate and solve a Pareto problem for this economy.
- Define a competitive equilibrium with sequential trading of a complete set of one-period Arrow securities.
- Does a competitive equilibrium with sequential trading of a complete set of one-period Arrow securities exist for this economy? If it does, compute it. If it does not, explain why.

Solution:

Convention. Utility terms on histories with subjective probability zero contribute nothing ($0 \cdot \ln 0 = 0$). Consumption there is only restricted by $c \geq 0$ and the budget.

(a) $\max \lambda \sum_t \sum_{s^t} \beta^t \ln c_t^1 \pi_t^1 + (1 - \lambda) \sum_t \sum_{s^t} \beta^t \ln c_t^2 \pi_t^2$ s.t. $c_t^1 + c_t^2 \leq 1$. At $t = 0$ both have probability one: $\lambda/c_0^1 = (1 - \lambda)/c_0^2$, so $c_0^1 = \lambda$. For $t \geq 1$ on history 1 only consumer 1's utility counts, so $c^1 = 1$, $c^2 = 0$. On history 2, $c^1 = 0$, $c^2 = 1$. This is the formula $c_t^1 = \lambda \pi_t^1 / (\lambda \pi_t^1 + (1 - \lambda) \pi_t^2)$ of 8.16(a). Each consumer gets everything in the history he believes in.

(b) As in 8.15(a): kernels, natural debt limits $A_t^i(s^t)$, allocation and portfolios with $\tilde{a}_0^i = 0$. Each consumer solves (8.8.3)–(8.8.4) under his own beliefs, and markets clear.

(c) **It exists, and it involves no trade.** The endowment gives consumer 1 .5 at $t = 0$, 1 forever on history 1 and 0 on history 2. That is exactly the Pareto allocation with $\lambda = .5$. Prices at $t = 0$: consumer 1's Euler equation on history 1 gives $\tilde{Q}(s_1=1) = \beta \cdot 1 \cdot \frac{.5}{1} = .5\beta$; consumer 2's on history 2 gives $\tilde{Q}(s_1=0) = .5\beta$; afterwards $\tilde{Q} = \beta$. At these prices consumer 1 is content with zero holdings of claims on history 1. He would like to *sell* claims on history 2 (positive price, zero value to him), but his natural debt limit there is $A^1(\text{history 2}) = 0$: his endowment is zero forever, so $\tilde{a}^1(0) \geq 0$ binds.

Symmetrically for consumer 2 on history 1. Nobody trades. The allocation is Pareto optimal. The natural debt limits *bind*, contrary to section 8.8.4: the Inada argument needs every history to have positive subjective probability.

Exercise 8.19 Diverse beliefs, V

Question

pp. 321–322. A pure exchange economy is populated by two consumers. Consumer i has preferences over history-contingent consumption sequences $\{c_t^i(s^t)\}$ that are ordered by $\sum_{t=0}^{\infty} \sum_{s^t} \beta^t u(c_t^i(s^t)) \pi_t^i(s^t)$, where $u(c) = \ln(c)$, $\beta \in (0, 1)$, and $\pi_t^i(s^t)$ is a density that consumer i assigns to history s^t . The state space is time invariant. In particular, $s_t \in S = \{0, .5, 1\}$ for all $t \geq 0$. Only two histories are possible for $t = 0, 1, 2, \dots$: history 1: $.5, 1, 1, 1, 1, \dots$; history 2: $.5, 0, 0, 0, 0, \dots$. Consumer 1 assigns probability 1 to history 1 and probability 0 to history 2, while consumer 2 assigns probability 0 to history 1 and probability 1 to history 2. Nature assigns equal probabilities to the two histories. The endowments of the two consumers are:

$$y_t^1 = 1 - s_t, \quad y_t^2 = s_t.$$

- Formulate and solve a Pareto problem for this economy.
- Define a competitive equilibrium with sequential trading of a complete set of one-period Arrow securities.
- Does a competitive equilibrium with sequential trading of a complete set of one-period Arrow securities exist for this economy? If it does, compute it. If it does not, explain why.

Solution:

(a) Preferences are those of 8.18, so the Pareto set is the same: $c_0^1 = \lambda$, $c^1 = 1$ on history 1 and 0 on history 2 ($t \geq 1$), $c^2 = 1 - c^1$.

(b) As in 8.18(b).

(c) **It exists, with maximal trade.** Now each consumer owns everything in the history he rules out: consumer 1 has 1 forever on history 2 and 0 on history 1. Natural debt limits: $A^1(\text{history 2}) = A^2(\text{history 1}) = 1/(1 - \beta)$ (claims priced at $\tilde{Q} = \beta$ after $t = 1$), and $A^1(\text{history 1}) = A^2(\text{history 2}) = 0$. Guess symmetric prices $\tilde{Q}(s_1=1) = \tilde{Q}(s_1=0) = q$. Consumer 1 sells history-2 claims up to his limit, $\tilde{a}^1(0) = -1/(1 - \beta)$, and uses the proceeds to buy $\tilde{a}^1(1) = 1/(1 - \beta)$. The cost is $q[1/(1 - \beta) - 1/(1 - \beta)] = 0$, so $c_0^1 = .5$. His Euler equation on history 1: $q = \beta \cdot 1 \cdot \frac{c_0^1}{c^1(h_1)}$. With $c^1(h_1) = 1$ this gives $q = .5\beta$. For $t \geq 1$ on history 1: he holds $1/(1 - \beta)$, has endowment 0, and consumes 1, since $1 + \beta \cdot \frac{1}{1 - \beta} = 0 + \frac{1}{1 - \beta}$. On history 2 he owes $1/(1 - \beta)$, pays his whole endowment 1 each period, and consumes 0. Consumer 2 is symmetric; the markets clear. The allocation is the Pareto allocation with $\lambda = .5$, the same as in 8.18, but it is reached by trade in which each consumer borrows *exactly up to* his natural debt limit in the history he rules out.

Comparison. In both 8.18 and 8.19 an equilibrium exists because $c \geq 0$ is allowed and zero-probability histories carry no utility weight. What differs is who owns what. When the “impossible” history is realised, the losing consumer consumes zero forever, and his beliefs cannot be revised by Bayes’ rule. Equilibrium relies on debts that are enforced at the natural limit.

Numerical exercise — see the companion code

$\beta = .95$: $\tilde{Q}_0 = (.475, .475)$; the notebook verifies the $t = 0$ budget, the roll-over budgets on both histories, and that the debt limit binds.

Risk-Free Bonds, Entropy and Beliefs Again (8.20–8.23)

Exercise 8.20 Risk-free bonds

Question

pp. 322–323. An economy consists of a single representative consumer who ranks streams of a single nonstorable consumption good according to $\sum_{t=0}^{\infty} \sum_{s^t} \beta^t \ln[c_t(s^t)] \pi_t(s^t)$. Here $\pi_t(s^t)$ is the subjective probability that the consumer attaches to a history s^t of a Markov state s_t , where $s_t \in \{1, 2\}$. Assume that the subjective probability $\pi_t(s^t)$ equals the objective probability. Feasibility for this pure endowment economy is expressed by the condition $c_t \leq y_t$, where y_t is the endowment at time t . The endowment is exogenous and governed by

$$y_{t+1} = \lambda_{t+1} \lambda_t \cdots \lambda_1 y_0$$

for $t \geq 0$ where $y_0 > 0$. Here λ_t is a function of the Markov state s_t . Assume that $\lambda_t = 1$ when $s_t = 1$ and $\lambda_t = 1 + \zeta$ when $s_t = 2$, where $\zeta > 0$. States $s^t = [s_t, \dots, s_0]$ are known at time t , but future states are not. The state s_t is described by a time invariant Markov chain with initial probability distribution $\pi_0 = [1 \ 0]'$ and transition density defined by the stochastic matrix

$$P = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix}$$

where $P_{ij} = \text{Prob}[s_{t+1} = j | s_t = i]$ for $i = 1, 2$ and $j = 1, 2$. Assume that $P_{ij} \geq 0$ for all pairs (i, j) .

- a. Show how to deduce $\pi_t(s^t)$ from (π_0, P) .
- b. Define a competitive equilibrium with sequential trading of Arrow securities.
- c. Compute a competitive equilibrium with sequential trading of Arrow securities.
- d. Let p_t^b be the time t price of a risk-free claim to one unit of consumption at time $t + 1$. Define a competitive equilibrium with sequential trading of risk-free claims to consumption one period ahead.
- e. Let $R_t = (p_t^b)^{-1}$ be the one-period risk-free gross interest rate. Give a formula for R_t and tell how it depends on the history s^t .
- f. Suppose that $\beta = .95$, $\zeta = .02$ and $P = \begin{bmatrix} 1 & 0 \\ .5 & .5 \end{bmatrix}$. Please compute R_t when $s_t = 1$. Then compute R_t when $s_t = 2$.
- g. What parts of your answers depend on assuming that the subjective probability $\pi_t(s^t)$ equals the objective probability?

Solution:

- (a) $\pi_t(s^t) = \pi_0(s_0) P_{s_0 s_1} P_{s_1 s_2} \cdots P_{s_{t-1} s_t}$; $\pi_0 = (1, 0)$ means $s_0 = 1$.
- (b) A kernel $\tilde{Q}_t(s_{t+1} | s^t)$, natural debt limits $A_t(s^t)$, an allocation $c_t(s^t)$ and portfolios $\tilde{a}_{t+1}(s_{t+1}, s^t)$ with $\tilde{a}_0 = 0$. Given the kernel, the consumer maximises utility subject to (8.8.3)–(8.8.4). Markets clear: $c_t = y_t$, $\tilde{a}_{t+1} = 0$ (zero net supply).

(c) $c_t = y_t$, $\tilde{a} \equiv 0$, and from (8.8.6) with $u'(c) = 1/c$:

$$\tilde{Q}_t(s_{t+1}|s^t) = \beta P_{s_t s_{t+1}} \frac{y_t}{y_{t+1}} = \frac{\beta P_{s_t s_{t+1}}}{\lambda(s_{t+1})} \equiv Q(s_{t+1}|s_t).$$

Natural debt limits: with log utility $q_\tau^t y_\tau = \beta^{\tau-t} \pi y_t$, so $A_t = y_t/(1 - \beta)$.

(d) A price $p_t^b(s^t)$, an allocation, and bond holdings b_{t+1} such that the consumer maximises utility s.t. $c_t + p_t^b b_{t+1} \leq y_t + b_t$, $b_0 = 0$, and a debt limit that rules out Ponzi schemes; and markets clear: $c_t = y_t$, $b_{t+1} = 0$.

(e) The Euler equation at $b = 0$: $p_t^b = \beta E_t[y_t/y_{t+1}] = \sum_{s'} Q(s'|s_t)$, so

$$R_t = \left[\beta \sum_j \frac{P_{s_t j}}{\lambda(j)} \right]^{-1},$$

which depends on the history only through the current state s_t .

(f) $s_t = 1$: $R = 1/(.95 \cdot 1) = 1.052632$. $s_t = 2$: $R = 1/ [.95(.5 + .5/1.02)] = 1.063054$. With $\pi_0 = (1, 0)$ and $P_{11} = 1$, state 2 is never visited, so $R(2)$ is an off-path value. The economy stays in state 1 with zero growth and $R = 1/\beta$.

(g) None of the equilibrium conditions uses the objective probabilities. The allocation ($c = y$) does not depend on probabilities at all. Prices Q , p^b and R are computed from the consumer's *subjective* P ; with a different subjective matrix the same formulas hold with that matrix. What depends on subjective = objective is the interpretation of prices as expectations under the data-generating process: average realised returns, the frequency with which $R(2)$ is observed, and statistical tests of asset-pricing Euler equations.

Exercise 8.21 Entropy

Question

p. 323. Let $\tilde{f}(s)$ and $f(s)$ be two alternative probability density functions for a random variable s . Define the relative entropy of f with respect to $\tilde{f}$ by

$$\text{ent}(f, \tilde{f}) = \int \log \left(\frac{\tilde{f}(s)}{f(s)} \right) \tilde{f}(s) ds = \int \log \left(\frac{\tilde{f}(s)}{f(s)} \right) \left(\frac{\tilde{f}(s)}{f(s)} \right) f(s) ds.$$

Prove that $\text{ent}(f, \tilde{f}) \geq 0$ and that $\text{ent}(\tilde{f}, \tilde{f}) = 0$.

Hint 1: Define $m(s) = \left(\frac{\tilde{f}(s)}{f(s)} \right)$ and express entropy as $\int m(s) \log m(s) f(s) ds$.

Hint 2: Verify that $m \log m \geq m - 1$.

Hint 3: Verify that $\int m(s) f(s) = 1$.

Solution:

Assume $\tilde{f} \ll f$ ($f > 0$ wherever $\tilde{f} > 0$), so $m = \tilde{f}/f$ is well defined; set $0 \log 0 = 0$.

Hint 1. $\text{ent}(f, \tilde{f}) = \int \log m \cdot m f ds = \int m \log m f ds$.

Hint 2. Let $g(m) = m \log m - m + 1$ on $m \geq 0$. Then $g(1) = 0$, $g'(m) = \log m$ (< 0 for $m < 1$, > 0 for $m > 1$), and $g''(m) = 1/m > 0$. So g has its unique minimum 0 at $m = 1$, and $g \geq 0$; at $m = 0$, $g(0) = 1 > 0$.

Hint 3. $\int m f ds = \int \tilde{f} ds = 1$.

Proof. $\text{ent}(f, \tilde{f}) = \int m \log m f ds \geq \int (m - 1) f ds = 1 - 1 = 0$. Equality requires $g(m(s)) = 0$ f -almost everywhere, i.e. $\tilde{f} = f$ a.e. For $\tilde{f} = f$, $m \equiv 1$ and $\log 1 = 0$, so $\text{ent}(f, f) = 0$.

Numerical exercise — see the companion code

$m \log m \geq m - 1$ on a grid of 2×10^5 points in $(0, 50]$; $\text{ent} \geq 0$ and $\int m f = 1$ on 1000 random pairs of discrete densities.

Exercise 8.22 Heterogeneous beliefs again

Question

pp. 323–324. Consider an example of the heterogeneous beliefs economy described in Appendix B of this chapter. Suppose that $S = \{0, 1\}$, $I = 2$, and that agent i believes that s_t is independently and identically distributed with $\text{Prob}(s_t = 0) = \alpha_i$. Assume that $u_i(c_t^i) = \log c_t^i$ for both consumers. The aggregate endowment is $y_t(s^t) = 1 - .5s_t$. Feasibility requires $c_t^1 + c_t^2 \leq y_t(s^t)$. A time 0 Pareto planner puts weight $\lambda \in (0, 1)$ on consumer 1 and weight $(1 - \lambda)$ on consumer 2. Suppose that $\{s_t\}_{t=1}^\infty$ is truly independently and identically distributed with $\text{Prob}(s_t = 0) = \tilde{\alpha}$. Assume that $I = 2$, $\alpha_2 = \tilde{\alpha}$, $\alpha_1 \neq \tilde{\alpha}$, so that consumer 2 knows the true distribution, while consumer 1 does not.

a. Let $L_t^2(s^t) = \frac{\pi_t^2(s^t)}{\pi_t^1(s^t)}$. Show that a Pareto optimal allocation satisfies

$$c_t^2(s^t) = \left[\frac{(1 - \lambda)L^2(s^t)}{\lambda + (1 - \lambda)L_t^2(s^t)} \right] y_t(s^t).$$

b. Find a formula for the price $\tilde{Q}(s_{t+1}|s^t)$ of Arrow securities.

c. State assumptions under which you can show that

$$\tilde{Q}_t(s_{t+1}|s^t) \rightarrow \beta \left(\frac{y_t(s_t)}{y_{t+1}(s_{t+1})} \right) \pi_t^2(s_{t+1}|s^t),$$

as $t \rightarrow +\infty$ in some sense. Describe the sense in which what you claim is true.

d. Use your findings from part **c** to argue that the “tail” of the economy resembles a representative consumer economy.

Solution:

(a) FOC (8.B.2) with $\lambda_1 = \lambda$, $\lambda_2 = 1 - \lambda$, $u' = 1/c$: $\lambda \beta^t \pi_t^1 / c_t^1 = (1 - \lambda) \beta^t \pi_t^2 / c_t^2$, so

$c_t^1 = c_t^2 \lambda / [(1 - \lambda)L_t]$. Feasibility $c_t^1 + c_t^2 = y_t$ gives $c_t^2 [1 + \frac{\lambda}{(1-\lambda)L_t}] = y_t$, which is the stated formula. ✓

(b) Use consumer 2's Euler equation (8.8.6), $\tilde{Q} = \beta \pi^2(s') c_t^2 / c_{t+1}^2$, with $L_{t+1} = L_t \ell(s')$, $\ell(s') = \pi^2(s') / \pi^1(s')$ (i.i.d. beliefs):

$$\frac{c_t^2}{c_{t+1}^2} = \frac{y_t}{y_{t+1}} \frac{L_t}{L_{t+1}} \frac{\lambda + (1 - \lambda)L_{t+1}}{\lambda + (1 - \lambda)L_t}$$

so that

$$\tilde{Q}(s_{t+1}|s^t) = \beta \frac{y_t}{y_{t+1}} \frac{\lambda \pi^1(s_{t+1}) + (1 - \lambda)L_t \pi^2(s_{t+1})}{\lambda + (1 - \lambda)L_t}$$

using $\pi^2(s')L_t/L_{t+1} = \pi^1(s')$. Prices use a consumption-share-weighted average of beliefs, with weight $w_t = \lambda/(\lambda + (1 - \lambda)L_t) = c_t^1/y_t$ on consumer 1's beliefs.

(c) Assumptions: s_t is i.i.d. under the truth with $\alpha_2 = \tilde{\alpha} \in (0, 1)$; $\alpha_1 \in (0, 1)$ and $\alpha_1 \neq \tilde{\alpha}$ (so both consumers give positive probability to both states and $\log \ell$ is finite); $\lambda \in (0, 1)$. Then $\log L_t = \sum_{\tau} \log \ell(s_{\tau})$ is a sum of i.i.d. terms with true mean $\text{ent}(\pi^2, \pi^1) = \sum_s \pi^2(s) \log[\pi^2(s)/\pi^1(s)] > 0$ (8.21, strict since $\pi^1 \neq \pi^2$). By the strong law of large numbers $\frac{1}{t} \log L_t \rightarrow \text{ent}(\pi^2, \pi^1) > 0$ almost surely, so $L_t \rightarrow \infty$ and $w_t \rightarrow 0$ a.s. Hence, for each s_{t+1} ,

$$\tilde{Q}(s_{t+1}|s^t) - \beta \frac{y_t}{y_{t+1}} \pi^2(s_{t+1}) = \beta \frac{y_t}{y_{t+1}} w_t [\pi^1(s_{t+1}) - \pi^2(s_{t+1})] \rightarrow 0$$

almost surely under the true probability measure: along almost every sample path, not uniformly, and not along the zero-probability paths on which consumer 1's beliefs look better. The gap shrinks geometrically, $w_t \approx \frac{\lambda}{1-\lambda} e^{-t \text{ent}(\pi^2, \pi^1)}$.

(d) Along almost every path $c_t^1/y_t = w_t \rightarrow 0$ and $c_t^2/y_t \rightarrow 1$. The Arrow prices converge to $\beta \pi^2(s') y_t / y_{t+1}$, the kernel of a representative consumer with log utility and *correct* beliefs who consumes the aggregate endowment. So the tail of the economy is a representative-agent, rational-expectations economy: the market selects for accurate beliefs (Friedman's conjecture, Blume–Easley).

Numerical exercise — see the companion code

Illustration $\tilde{\alpha} = \alpha_2 = .5$, $\alpha_1 = .7$, $y = (1, .5)$, $\lambda = .5$, $\beta = .95$, five simulated paths of 3000 periods: the formula in (b) equals consumer 2's Euler equation, $\frac{1}{t} \log L_t$ is within .03 of $\text{ent}(\pi^2, \pi^1) = .08718$, and the price gap at $t = 3000$ is below 10^{-10} (figure).

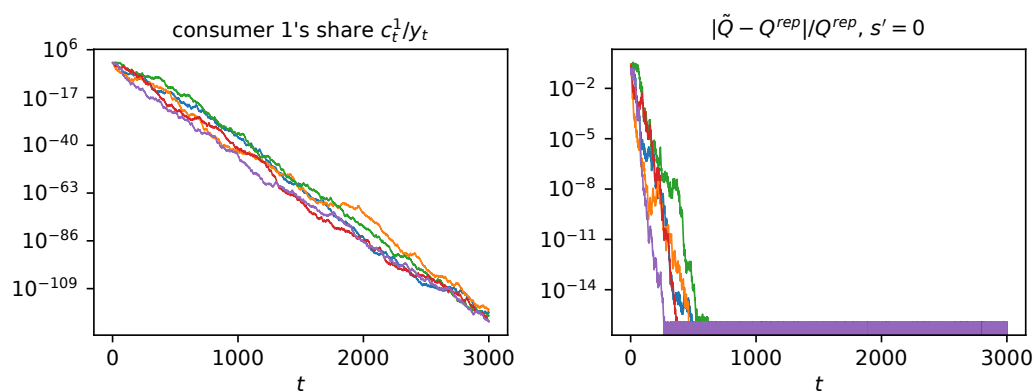


Figure 2: Exercise 8.22: consumer 1's consumption share and the relative gap between Arrow prices and the representative-agent kernel, five sample paths.

Exercise 8.23 Recursive version of Pareto problem

Question

pp. 324–325. **a.** Please reread section 8.12.

b. Assume exactly the same setting as in section 8.12, except now assume that consumer 1 believes probabilities Π_s^1 and that consumer two believes probabilities Π_s^2 .

c. Formulate a recursive version of the Pareto problem in terms of the Bellman equation

$$P(v) = \max_{\{c_s, w_s\}} \sum_{i=s}^S [u_2(1 - c_s) + \beta P(w_s)] \Pi_s^2$$

where the maximization is subject to

$$\sum_{s=1}^S [u_1(c_s) + \beta w_s] \Pi_s^1 \geq v$$

where c_s is consumption assigned to consumer 1, $1 - c_s$ is consumption assigned to consumer 2, v is the promised value assigned to consumer 1, w_s is a continuation value assigned to consumer 1, $P(v)$ is the value attained by consumer 2, and $P(w_s)$ is the continuation value assigned to consumer 2.

d. Assuming that $P(v)$ is concave and differentiable, find first-order conditions for c_s, w_s for $s = 1, \dots, S$.

e. Argue that these first-order conditions imply that for states i such that $\frac{\Pi_s^1}{\Pi_s^2} > 1$, the Pareto planner gives larger values of *both* c_s^1 and w_s^1 than he or she does in states for which $\frac{\Pi_s^1}{\Pi_s^2} < 1$.

f. How do the outcomes described in item **e** relate to the asymptotic outcomes described in Appendix B of this chapter?

Solution:

Misprints. In (c) the sum should run over $s = 1, \dots, S$ (not “ $i = s$ ”); in (e), “states i ” means states s , and c_s^1, w_s^1 are c_s, w_s .

(a)–(c) Setting of section 8.12: i.i.d. s , aggregate endowment 1, $c_s \in [0, 1]$, $w_s \in V$. Each consumer evaluates promises with his own probabilities:

$$P(v) = \max_{\{c_s, w_s\}} \sum_{s=1}^S \Pi_s^2 [u_2(1 - c_s) + \beta P(w_s)] \quad \text{s.t.} \quad \sum_{s=1}^S \Pi_s^1 [u_1(c_s) + \beta w_s] \geq v.$$

(d) $\mathcal{L} = \sum_s \Pi_s^2 [u_2(1 - c_s) + \beta P(w_s)] + \theta \left(\sum_s \Pi_s^1 [u_1(c_s) + \beta w_s] - v \right)$. FOCs:

$$c_s : -\Pi_s^2 u_2'(1 - c_s) + \theta \Pi_s^1 u_1'(c_s) = 0 \iff \frac{u_2'(1 - c_s)}{u_1'(c_s)} = \theta \frac{\Pi_s^1}{\Pi_s^2},$$

$$w_s : \beta \Pi_s^2 P'(w_s) + \theta \beta \Pi_s^1 = 0 \iff P'(w_s) = -\theta \frac{\Pi_s^1}{\Pi_s^2}.$$

Envelope: $P'(v) = -\theta$. So $P'(w_s) = P'(v) \Pi_s^1/\Pi_s^2$. With common beliefs this collapses to (8.12.3), $w_s = v$.

(e) Let $r_s = \Pi_s^1/\Pi_s^2$. *Continuation values:* $-P'(w_s) = \theta r_s$. P is concave, so $-P'$ is nondecreasing (strictly increasing if P is strictly concave). Hence $r_s > 1 \Rightarrow -P'(w_s) > -P'(v) \Rightarrow w_s > v$, and $r_s < 1 \Rightarrow w_s < v$. *Consumption:* $g(c) = u'_2(1-c)/u'_1(c)$ is strictly increasing in c (the numerator rises because $u''_2 < 0$; the denominator falls because $u''_1 < 0$). So $c_s = g^{-1}(\theta r_s)$ is increasing in r_s . Therefore states with $r_s > 1$ get larger c_s and larger w_s than states with $r_s < 1$. The planner bets on consumer 1's behalf where he is relatively optimistic, both today and in promised utility.

(f) The relative weight $\theta_t = -P'(v_t)$ is the state variable, and it evolves as $\theta_{t+1} = \theta_t r_{s_{t+1}}$:

$$\theta_t = \theta_0 \prod_{\tau} \frac{\Pi_{s_\tau}^1}{\Pi_{s_\tau}^2} = \theta_0 \frac{\pi_t^1(s^t)}{\pi_t^2(s^t)}.$$

These are exactly the continuation Pareto weights $\lambda_i^t(s^t) = [\pi_t^i/\pi_t^1]\lambda_i$ of Appendix B (as a ratio). If consumer 2's beliefs are the truth, $\frac{1}{t} \log \theta_t \rightarrow -\text{ent}(\Pi^2, \Pi^1) < 0$ a.s. Then $\theta_t \rightarrow 0$: consumer 1's promised value v_t drifts to the lower end of V and his consumption to zero (8.B.4). If consumer 1 is right, the reverse happens. If neither is right, the one with the smaller relative entropy wins. The state-by-state tilting in (e) is the one-step mechanism behind the asymptotic result: on average, true frequencies move w towards the better-informed consumer.

Numerical exercise — see the companion code

Illustration: $u_1 = \log c$, $u_2 = \log(1-c)$, $\Pi^1 = (.3, .7)$, $\Pi^2 = (.5, .5)$, $\beta = .9$. Then $c_s = \theta r_s/(1 + \theta r_s)$ and $w_s = v(\theta r_s)$. $v(\theta)$ and $P(\theta)$ are computed as truncated sums over binomial counts of states (350 dates), independently of the Bellman equation. Checks at $\theta \in \{.5, 1, 3\}$: both Bellman equations hold; the finite-difference slope $dP/dv = -\theta$; $w_s > v$ iff $r_s > 1$. At $\theta = 1$: $v = -5.3255$, $P = -5.3658$, $c = (.375, .5833)$, $w = (-7.4258, -4.2048)$.

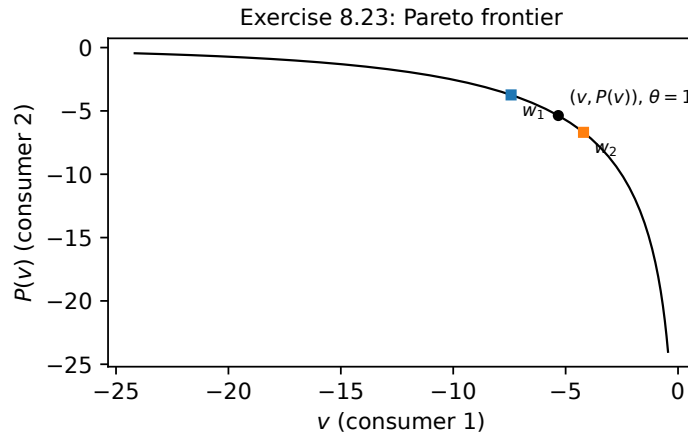


Figure 3: Exercise 8.23: Pareto frontier with heterogeneous beliefs; continuation values $w_1 < v < w_2$ at $\theta = 1$ ($r_1 = .6 < 1 < r_2 = 1.4$).

Recovering Probabilities from Prices, and Corners Again (8.24–8.28)

Exercise 8.24 Recovering probabilities from prices

Question

pp. 325–326. Consider a representative agent economy with an exogenous endowment of a single consumption good governed by a two-state Markov chain with state $s_t \in \{1, 2\}$ having transition matrix P with typical element $P_{ij} = \text{Prob}(s_{t+1} = j | s_t = i)$. The representative consumer's endowment y_t is a time invariant function $y(s_t) > 0$ of the Markov state. So is his consumption. The representative consumer orders consumption streams by

$$E_0 \sum_{t=0}^{\infty} \beta^t u(c_t),$$

where $\beta \in (0, 1)$ and $u(c) = \frac{1}{1-\gamma} c^{1-\gamma}$ for $\gamma > 0$. At each date $t \geq 0$, there are complete markets in h -step ahead Arrow securities of horizons $h = 1, 2$. Recall formula (8.10.3) for the h -step ahead Arrow securities prices, which for convenience we repeat here

$$Q_h(s_{t+h} | s_t) = \sum_{s_{t+1}} Q_1(s_{t+1} | s_t) Q_{h-1}(s_{t+h} | s_{t+1}).$$

- With an abuse of notation, arrange the one-period Arrow security prices into a 2×2 matrix Q defined according to $Q_{ij} = Q(s_{t+1} = j | s_t = i)$. Then show that formula (8.10.3) can be represented as $Q^h = Q Q^{h-1}$ for $h \geq 2$.
- Suppose that you observe a complete set of one and two period ahead Arrow securities prices for *one* and only one state i . That is, you know the i th row of both Q and Q^2 . Show how you can recover both rows of Q from this information.
- Recall that consumption of a representative consumer is a time-invariant function of the Markov state. Find a formula for each element of Q in terms of β, γ , the consumption rates in the two states, and P .
- Prove that if you know all elements of Q , you can infer β and all elements of P even if you don't know γ .

Solution:

(a) Let Q_h be the matrix with $(Q_h)_{ij} = Q_h(j | i)$. (8.10.3) reads $(Q_h)_{ij} = \sum_k Q_{ik}(Q_{h-1})_{kj} = (Q Q_{h-1})_{ij}$. With $Q_1 = Q$, induction gives $Q_h = Q Q_{h-1} = Q^h$.

(b) Take $i = 1$ ($i = 2$ is symmetric). Row 1 of Q^2 is $Q_{1j}^2 = Q_{11}Q_{1j} + Q_{12}Q_{2j}$, so

$$Q_{2j} = \frac{Q_{1j}^2 - Q_{11}Q_{1j}}{Q_{12}}, \quad j = 1, 2,$$

provided $Q_{12} > 0$, i.e. $P_{12} > 0$ (state 2 is reachable from state 1).

(c) From (8.9.12) with $c = y$:

$$Q_{ij} = \beta P_{ij} \left(\frac{y(j)}{y(i)} \right)^{-\gamma}.$$

(d) Let $D = \text{diag}(y(1)^{-\gamma}, y(2)^{-\gamma})$. Then $Q = \beta D^{-1} P D$, so Q is *similar* to βP and has the same eigenvalues. P is stochastic, with eigenvalues 1 and $P_{11} + P_{22} - 1 \in (-1, 1)$. So Q has eigenvalues β and $\beta(P_{11} + P_{22} - 1)$, and

$$\beta = \text{the largest (Perron) eigenvalue of } Q = \frac{1}{2} \left[\text{trace } Q + \sqrt{(\text{trace } Q)^2 - 4 \det Q} \right].$$

Its eigenvector: $Q(D^{-1}\mathbf{1}) = \beta D^{-1}P\mathbf{1} = \beta D^{-1}\mathbf{1}$, so $z \propto D^{-1}\mathbf{1} = (y(1)^\gamma, y(2)^\gamma)'$. The Perron vector is unique up to scale. Then $P = \beta^{-1} D Q D^{-1}$, i.e.

$$P_{ij} = \frac{Q_{ij} z_j}{\beta z_i},$$

which uses only ratios $z_j/z_i = (y(j)/y(i))^\gamma$, which the eigenvector reveals. We never need γ (nor the y 's) separately. This is Ross's recovery theorem in a two-state economy.

Numerical exercise — see the companion code

$\beta = .95$, $\gamma = 3$, $y = (1, 1.1)$, $P = \begin{bmatrix} .9 & .1 \\ .4 & .6 \end{bmatrix}$; $Q = \begin{bmatrix} .855 & .071375 \\ .505780 & .570 \end{bmatrix}$. The Perron recovery returns $\beta = .95$ and P exactly, with $z_2/z_1 = 1.1^3$; (b) recovers row 2 from row 1 of Q and Q^2 .

Exercise 8.25 Recovering probabilities from prices, II

Question

pp. 326–327. Consider a representative agent economy with an exogenous endowment of a single consumption good whose geometric growth rate is governed by a two-state Markov chain with state $s_t \in \{1, 2\}$ having transition matrix P with typical element $P_{ij} = \text{Prob}(s_{t+1} = j | s_t = i)$. The growth rate of the consumer's endowment y_t is a time invariant function $\lambda(s_t) > 0$ of the Markov state. The endowment obeys $y_{t+1} = \lambda_{t+1}y_t$ for $t \geq 0$. The representative consumer orders consumption streams by $E_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$, where $\beta \in (0, 1)$ and $u(c) = \frac{1}{1-\gamma} c^{1-\gamma}$ for $\gamma > 0$. There are complete markets in one and two period Arrow securities.

- a. With an abuse of notation, arrange the one-period Arrow security prices into a 2×2 matrix Q defined according to $Q_{ij} = Q(s_{t+1} = j | s_t = i)$. Then show that formula (8.10.3) can be represented as $Q^h = QQ^{h-1}$ for $h \geq 2$.
- b. Suppose that you observe all one and two period Arrow securities prices for a single state i . That is, you know the i th row of both Q and Q^2 . Show how you can recover both rows of Q from this information.
- c. Find a formula for each element of Q in terms of β, γ , and P .
- d. Prove that if you know all elements of Q , you *cannot* infer β and all elements of P even if you know γ .

Solution:

(a)–(b) Identical to 8.24(a)–(b): these steps use only (8.10.3).

(c) $u'(c_{t+1})/u'(c_t) = \lambda(s_{t+1})^{-\gamma}$, so

$$Q_{ij} = \beta P_{ij} \lambda(j)^{-\gamma}, \quad Q = \beta P \Lambda, \quad \Lambda = \text{diag}(\lambda(1)^{-\gamma}, \lambda(2)^{-\gamma}).$$

(The formula also needs the growth rates $\lambda(j)$.)

(d) *Reading:* the growth rates $\lambda(s)$ are unknown to the observer (if they were known, β would equal the common row sum of $Q\Lambda^{-1}$, since $P\mathbf{1} = \mathbf{1}$). Since $P\mathbf{1} = \mathbf{1}$, $Q(\beta\Lambda)^{-1}\mathbf{1} = \mathbf{1}$, so $x \equiv Q^{-1}\mathbf{1}$ satisfies $x_j = 1/(\beta\lambda(j)^{-\gamma})$ and $P = Q \text{diag}(x)$. Thus P and the *products* $\beta\lambda(j)^{-\gamma}$ are identified, but β is not. For any $\beta' \in (0, 1)$ set $\lambda'(j)^{-\gamma} = 1/(\beta'x_j)$. Then $\beta'P\Lambda' = Q \text{diag}(x) \text{diag}(1/x) = Q$: the same prices. Knowing γ does not help, because only $\lambda(j)^{-\gamma}$ enters. Contrast with 8.24: there $Q = \beta D^{-1}PD$ is a *similarity*, so scaling D leaves Q unchanged and β is the Perron root. Here Λ multiplies only on the right, so a common scale of Λ is absorbed by β . (Strictly, the statement's claim “cannot infer β and all elements of P ” is true because of β ; P itself is identified.)

Numerical exercise — see the companion code

$\beta = .95$, $\gamma = 2$, $\bar{\lambda} = (.97, 1.03)$, $P = \begin{bmatrix} .8 & .2 \\ .1 & .9 \end{bmatrix}$: $Q \text{diag}(Q^{-1}\mathbf{1}) = P$ exactly. $\beta' = .90$ with $\lambda' = (.944129, 1.002528)$, and $\beta' = .99$ with $\lambda' = (.990211, 1.051461)$, reproduce Q to machine precision.

Exercise 8.26 Recovering probabilities from prices, III

Question

p. 327. Consider a representative agent economy with an exogenous endowment of a single consumption good governed by a two-state Markov chain with state $s_t \in \{1, 2\}$ having transition matrix P with typical element $P_{ij} = \text{Prob}(s_{t+1} = j | s_t = i)$. The representative consumer's endowment y_t is a time invariant function $y(s_t) > 0$ of the Markov state. So is his consumption $c_t = c(s_t)$. The representative consumer orders consumption streams by $E_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$, where $\beta \in (0, 1)$, $u(c) = \frac{1}{1-\gamma} c^{1-\gamma}$ for $\gamma > 0$, and E_0 denotes a mathematical expectation conditioned on date 0 information, in this case the time 0 value s_0 of the state. At each date $t \geq 0$, there are complete markets in one-step ahead Arrow securities. There are also markets in h -period risk-free pure discount bonds. A risk-free pure discount h -period bond issued at t promises to pay one unit of consumption at date $t+h$ for sure. At time t , the equilibrium price of a sure claim on one unit of consumption at time $t+h$ is p_{t+h}^t .

- a. Please present a formula for a stochastic discount factor for pricing h -period ahead risky claims for this economy. Is this stochastic discount factor unique?
- b. Please find formulas for the equilibrium price p_{t+h}^t of h -period risk-free pure discount bonds for $h \geq 1$.
- c. At a single date t , you observe the state of the economy s_t . You also observe p_{t+h}^t for $h = 1, 2, 3, 4$. Your friend tells you that from these four observations you can infer β , P_{11} , P_{22} , and $\left(\frac{c(1)}{c(2)}\right)^{-\gamma}$ for this economy. Do you agree? Provide an argument.

Solution:

Let $\kappa = (c(1)/c(2))^{-\gamma}$. Then $Q = \beta \begin{bmatrix} P_{11} & P_{12}\kappa^{-1} \\ P_{21}\kappa & P_{22} \end{bmatrix}$.

(a) $m_{t,t+h} = \beta^h (c(s_{t+h})/c(s_t))^{-\gamma}$. A claim paying $x(s_{t+h})$ costs $E_t[m_{t,t+h}x] = \sum_j Q_{s_t,j}^h x(j)$. Given the probability measure P and complete Arrow markets, the SDF is *unique*: $m(j|i) = Q_{ij}^h / (P^h)_{ij}$ is pinned down by the Arrow prices. The decomposition is not unique, though. For any other transition matrix $\tilde{P}$ with the same zero pattern, $\tilde{m} = Q_{ij}^h / (\tilde{P}^h)_{ij}$ prices every claim identically. Prices reveal Q , i.e. probability times SDF, not the two factors separately.

(b) $p_{t+h}^t = \sum_j Q_{s_t,j}^h = e'_{s_t} Q^h \mathbf{1}$. E.g. $p_{t+1}^t = \beta(P_{11} + P_{12}/\kappa)$ if $s_t = 1$.

(c) **No.** By Cayley–Hamilton, $Q^2 = \tau Q - \delta I$ with $\tau = \text{trace } Q$ and $\delta = \det Q$. So the bond prices $p_h \equiv p_{t+h}^t$ (with $p_0 = 1$) satisfy

$$p_{h+2} = \tau p_{h+1} - \delta p_h, \quad h \geq 0.$$

Three of these relations involve $p_1, \dots, p_4$. The first two determine (τ, δ) , provided $p_2 \neq p_1^2$, and the third is then automatically satisfied: p_4 carries *no* new information. The data amount to three numbers, (p_1, τ, δ) , for four unknowns. The diagonal of Q has no κ , and in $\det Q$ the κ 's cancel:

$$\tau = \beta(P_{11} + P_{22}), \quad \delta = \beta^2(P_{11}P_{22} - P_{12}P_{21}) = \beta^2(P_{11} + P_{22} - 1).$$

So β solves $\beta^2 - \tau\beta + \delta = 0$: it is the larger root (the Perron root, as in 8.24), and $\sigma = P_{11} + P_{22} = \tau/\beta$ follows. The remaining datum, $p_1 = \beta(P_{11} + (1 - P_{11})/\kappa)$ at $s_t = 1$, is one equation in the two unknowns (P_{11}, κ) (with $P_{22} = \sigma - P_{11}$): a one-dimensional family of observationally equivalent economies. One can infer β and $P_{11} + P_{22}$, but not P_{11} , P_{22} and κ separately. Compare 8.28: there the two-period *Arrow* prices add the missing state-contingent information.

Numerical exercise — see the companion code

$\beta = .95$, $P_{11} = .9$, $P_{22} = .6$, $\kappa = 1.2$ at $s_t = 1$: $p = (.934167, .879937, .832368, .789053)$. The recursion holds; (τ, δ) from p_1, p_2, p_3 give $\beta = .95$. The economy with $P_{11} = .8$, $P_{22} = .7$, $\kappa = 1.090909$ produces the same four prices.

Exercise 8.27 Corners

Question

pp. 328–329. Consider an economy with exogenous endowments governed by a Markov chain with matrix $P = \begin{bmatrix} \lambda & (1-\lambda) \\ 1-\delta & \delta \end{bmatrix}$ that governs transitions of a state $s_t \in \{\bar{s}_1, \bar{s}_2\}$, where $\lambda \in (0, 1), \delta \in (0, 1)$. The initial probability distribution of s_0 is π_0 . The Markov chain induces a sequence $\{\pi_t(s^t)\}_{t=0}^\infty$ of distributions over histories $s^t = [s_t, s_{t-1}, \dots, s_0]$. There are equal numbers of two types of consumers. A consumer of type i receives an exogenous endowment $y^i(s_t) \geq 0$ of a nonstorable consumption good at time, Markov state t, s_t . Consumers of type 1 order consumption streams of the one good according to

$$V_1 = \sum_{t=0}^{\infty} \sum_{s^t} \beta^t c_t^1(s^t) \pi_t(s^t)$$

and consumers of type 2 order consumption streams according to

$$V_2 = \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \ln(c_t^2(s^t)) \pi_t(s^t)$$

where $c_t^i(s^t) \geq 0$ is the consumption of a type i consumer and $\beta \in (0, 1)$ is a common discount factor. Please note the nonnegativity constraint on consumption of each person (the force of this is that c_t^i is *consumption*, not *production*). The consumption good is tradable but nonstorable. There are equal numbers of the two types of consumer.

a. A Pareto planner chooses a consumption allocation to maximize the criterion $\alpha V_1 + (1 - \alpha)V_2$, where $\alpha \in (0, 1)$. Maximization is subject to the feasibility restrictions $\sum_i c_t^i(s^t) \leq \sum_i y^i(s_t)$ for all $t \geq 0$ for all s^t . Please characterize a Pareto optimal allocation and get as far as you can in computing it. Under what circumstances is the constraint $c_t^i(s^t) \geq 0$ binding? When it is binding, how is the optimal allocation related to the aggregate endowment $y^1(s_t) + y^2(s_t)$?

b. Please define a competitive equilibrium with trading each period $t \geq 0$ at each history s^t of a complete set of Arrow one-period-ahead state-contingent securities, being careful to include descriptions of all objects that comprise a competitive equilibrium.

c. Compute all objects that comprise a competitive equilibrium with trading each period $t \geq 0$ at each history s^t of a complete set of Arrow one-period-ahead state-contingent securities under the following alternative auxiliary assumptions:

Auxiliary Assumption 1: The equilibrium allocation is such that the restrictions $c_t^i(s^t) \geq 0$ never bind for any i .

Auxiliary Assumption 2: The equilibrium allocation is such that the restrictions $c_t^i(s^t) \geq 0$ always bind for some i .

Solution:

Write $Y(s) = y^1(s) + y^2(s)$.

(a) With multipliers $\zeta_t(s^t)$ on feasibility, the Kuhn–Tucker conditions are

$$\alpha\beta^t\pi_t \leq \zeta_t \quad (= \text{ if } c_t^1 > 0), \quad (1 - \alpha)\beta^t\pi_t/c_t^2 = \zeta_t.$$

If $c_t^1 > 0$, then $\zeta_t = \alpha\beta^t\pi_t$ and $c_t^2 = (1 - \alpha)/\alpha$. Type 2 never hits zero (Inada). Hence

$$c_t^2(s^t) = \min \left\{ Y(s_t), \frac{1 - \alpha}{\alpha} \right\}, \quad c_t^1(s^t) = \max \left\{ Y(s_t) - \frac{1 - \alpha}{\alpha}, 0 \right\}.$$

The allocation depends only on s_t . Type 1's constraint binds iff $Y(s_t) < (1 - \alpha)/\alpha$, i.e. in low-endowment states or when type 1's weight is small. Then type 2 consumes the whole aggregate endowment, $c^2 = Y(s_t)$, and consumer 2's consumption moves one-for-one with Y . Otherwise type 2 is fully insured at $(1 - \alpha)/\alpha$ and the risk-neutral type 1 absorbs all fluctuations.

(b) Objects: a kernel $Q(s'|s)$; natural debt limits $\bar{A}^i(s) = y^i(s) + \sum_{s'} Q(s'|s)\bar{A}^i(s')$; value functions $v^i(a, s)$; decision rules $c = h^i(a, s)$, $\hat{a}(s') = g^i(a, s, s')$; initial wealths $a_0^i = 0$. Given Q and $\bar{A}^i$, v^i solves $v^i(a, s) = \max\{u_i(c) + \beta \sum_{s'} v^i(\hat{a}(s'), s')P(s'|s)\}$ s.t. $c + \sum_{s'} Q(s'|s)\hat{a}(s') \leq y^i(s) + a$, $c \geq 0$, $-\hat{a}(s') \leq \bar{A}^i(s')$, with $u_1(c) = c$ and $u_2 = \ln$. Markets clear: $c^1 + c^2 = Y(s)$ and $\hat{a}^1(s') + \hat{a}^2(s') = 0$.

(c) *General solution.* Type 2 is always interior, so $Q(s'|s) = \beta P(s'|s)c^2(s)/c^2(s')$. Type 1's marginal value of wealth $\varphi(s) = \partial v^1/\partial a$ satisfies $\varphi(s) \geq 1$, with equality where $c^1(s) > 0$ (FOC for c), and the Euler equation $Q(s'|s)\varphi(s) = \beta P(s'|s)\varphi(s')$. Both hold with $\varphi(s) = \bar{c}/c^2(s)$ and

$$c^2(s) = \min\{Y(s), \bar{c}\}, \quad c^1(s) = Y(s) - c^2(s).$$

$\bar{c}$ comes from type 2's budget with $a_0^2 = 0$. Time-0 prices implied by Q are $q_t^0 = \beta^t \pi_t c^2(s_0)/c^2(s_t)$, so the value of c^2 is $c^2(s_0)/(1 - \beta)$ and the value of y^2 is $c^2(s_0) \sum_t \beta^t E_{s_0}[y^2/c^2]$:

$$e'_{s_0}(I - \beta P)^{-1} \left[\frac{y^2(s)}{\min\{Y(s), \bar{c}\}} \right]_s = \frac{1}{1 - \beta}. \quad (\dagger)$$

The left side is decreasing in $\bar{c}$, so $\bar{c}$ is unique. Holdings $\hat{a}^i(s) = \Upsilon^i(s) = [(I - Q)^{-1}(c^i - y^i)](s)$ and debt limits $\bar{A}^i = (I - Q)^{-1}y^i$ complete the equilibrium. The Pareto weight is $\alpha = 1/(1 + \bar{c})$.

Assumption 1 (never binds). $\varphi \equiv 1$, so $Q = \beta P$; $c^2 = \bar{c}$ constant, and $(\dagger)$ gives $\bar{c} = (1 - \beta)e'_{s_0}(I - \beta P)^{-1}y^2$, the annuity value of type 2's endowment. $c^1(s) = Y(s) - \bar{c}$, which requires $\bar{c} < \min_s Y(s)$. Type 2 is perfectly insured by the risk-neutral type 1 at actuarially fair prices.

Assumption 2 (binds). Only type 1 can be at the corner. *Reading:* “always bind” cannot mean $c^1 \equiv 0$ in every state, since type 1 would then never spend a positive wealth. It means the constraint binds in the low-endowment state $\bar{s}_1$ whenever it occurs, and

not in $\bar{s}_2$ (so $Y(\bar{s}_1) \leq \bar{c} < Y(\bar{s}_2)$). Then $c^1(\bar{s}_1) = 0$, $c^2(\bar{s}_1) = Y(\bar{s}_1)$, $c^2(\bar{s}_2) = \bar{c}$, $\varphi(\bar{s}_1) = \bar{c}/Y(\bar{s}_1) \geq 1$, and

$$Q = \beta \begin{bmatrix} \lambda & (1-\lambda)Y(\bar{s}_1)/\bar{c} \\ (1-\delta)\bar{c}/Y(\bar{s}_1) & \delta \end{bmatrix}, \quad \bar{c} = \frac{m_2 y^2(\bar{s}_2)}{\frac{1}{1-\beta} - m_1 y^2(\bar{s}_1)/Y(\bar{s}_1)},$$

where $(m_1, m_2) = e'_{s_0}(I - \beta P)^{-1}$. Claims paying in the corner state are expensive. Type 1 values wealth there at $\varphi > 1$, and the kernel is no longer βP .

Numerical exercise — see the companion code

Illustration $\lambda = .8$, $\delta = .6$, $\beta = .9$, $s_0 = \bar{s}_1$. *Case A* (Assumption 1): $y^1 = (1, 1.5)$, $y^2 = (.5, .3)$: $\bar{c} = .44375$ ($\alpha = .6926$), $Q = \beta P$, $\Upsilon^1 = (0, -.3125)$, $\bar{A}^1 = (11.406, 12.188)$, $\bar{A}^2 = (4.4375, 4.125)$. *Case B* (Assumption 2): $y^1 = (.05, 1.5)$, $y^2 = (.3, 1)$: $\bar{c} = .732558$ ($\alpha = .5772$), $c^1 = (0, 1.767442)$, $c^2 = (.35, .732558)$, $\varphi = (2.093, 1)$, $Q = \begin{bmatrix} .72 & .086 \\ .753488 & .54 \end{bmatrix}$, $\Upsilon^1 = (0, .581395)$. Both cases: Arrow budgets of both types hold in every state, type 1's Euler equation with φ holds, and the KKT signs are right.

Exercise 8.28 Recovering probabilities from prices, IV

Question

pp. 329–330. Consider an economy with an exogenous endowment governed by a Markov chain with transition matrix $P = \begin{bmatrix} \lambda & (1-\lambda) \\ 1-\delta & \delta \end{bmatrix}$ that governs transitions of a state $s_t \in \{\bar{s}_1, \bar{s}_2\}$, where $\lambda \in (0, 1)$, $\delta \in (0, 1)$. The initial distribution over s_0 is π_0 . The Markov chain induces a sequence $\{\pi_t(s^t)\}_{t=0}^\infty$ of distributions over histories $s^t = [s_t, s_{t-1}, \dots, s_0]$. There is a representative consumer who orders consumption plans according to

$$\sum_{t=0}^{\infty} \sum_{s^t} \beta^t u(c_t(s^t)) \pi_t(s^t)$$

where $u(c) = (1 - \gamma)^{-1} c^{1-\gamma}$ and $\gamma > 0$ and $\beta \in (0, 1)$. There is one consumer called the representative consumer. The representative consumer's endowment is $y_t = \bar{y}_1 > 0$ if $s_t = \bar{s}_1$ and $y_t = \bar{y}_2 > 0$ if $s_t = \bar{s}_2$.

a. Define a competitive equilibrium with trading at each date, history pair t, s^t of one-period-ahead Arrow state contingent securities. Please define all components of a competitive equilibrium, including the allocation and the price system.

b. Show that equilibrium one-period Arrow state-contingent security prices can be arranged into a (2×2) matrix $Q^{(1)}$ whose (i, j) component $Q_{i,j}^{(1)}$ is the price of one unit of consumption when Markov state $s_{t+1} = \bar{s}_j$ tomorrow when the Markov state s_t today is in state $\bar{s}_i$. Please give formulas for all elements of $Q^{(1)}$ in terms of the fundamental parameters of the economy $\beta, \gamma, \lambda, \delta, \bar{y}_1, \bar{y}_2$.

c. Define a competitive equilibrium with trading at each date, history pair t, s^t of both one-period-ahead and two-period-ahead Arrow state contingent securities. Let two-period Arrow state-contingent security prices be arranged into a (2×2) matrix $Q^{(2)}$ whose (i, j) component $Q_{i,j}^{(2)}$ is the price of one unit of consumption when Markov state $s_{t+2} = \bar{s}_j$ two periods ahead when the Markov state s_t today is in state $\bar{s}_i$. Please give formulas for all elements of $Q^{(2)}$ in terms of the fundamental parameters of the economy $\beta, \gamma, \lambda, \delta, \bar{y}_1, \bar{y}_2$.

d. (“inverse problem”) An outsider observes this economy. The outsider knows the theoretical structure of the economy but does not know the parameter values $[\lambda, \delta, \beta, \gamma, \bar{y}_1, \bar{y}_2]$. But at a particular date at which $s_t = \bar{s}_1$, the outsider observes

$$[Q^{(1)}(1, 1), Q^{(1)}(1, 2), Q^{(2)}(1, 1), Q^{(2)}(1, 2)].$$

Please interpret these observations. From these observations alone, can the outsider infer λ, δ, β ? Please explain your answer.

Solution:

(a) A kernel $Q(s'|s)$, natural debt limits $\bar{A}(s)$, a value function $v(a, s)$, decision rules $c = h(a, s)$, $\hat{a}(s') = g(a, s, s')$, and $a_0 = 0$. Given Q and $\bar{A}$, these solve the consumer's Bellman equation (8.9.8)–(8.9.10), and markets clear: $c = y(s)$, $\hat{a}(s') = 0$. The alloca-

tion is $c_t = y(s_t)$.

(b) From (8.9.12), $Q_{ij}^{(1)} = \beta P_{ij}(\bar{y}_j/\bar{y}_i)^{-\gamma}$. With $\kappa \equiv (\bar{y}_2/\bar{y}_1)^{-\gamma}$:

$$Q^{(1)} = \beta \begin{bmatrix} \lambda & (1-\lambda)\kappa \\ (1-\delta)/\kappa & \delta \end{bmatrix}.$$

(c) With both markets, the budget is (8.10.1) with $j \leq 2$. The equilibrium adds the two-period prices, the allocation is still $c = y$, and by (8.10.3) $Q^{(2)} = Q^{(1)}Q^{(1)}$:

$$Q_{11}^{(2)} = \beta^2[\lambda^2 + (1-\lambda)(1-\delta)], \quad Q_{12}^{(2)} = \beta^2(1-\lambda)(\lambda + \delta)\kappa,$$

$$Q_{21}^{(2)} = \beta^2(1-\delta)(\lambda + \delta)/\kappa, \quad Q_{22}^{(2)} = \beta^2[\delta^2 + (1-\lambda)(1-\delta)].$$

(For example, $Q_{12}^{(2)} = \beta\lambda \cdot \beta(1-\lambda)\kappa + \beta(1-\lambda)\kappa \cdot \beta\delta$.)

(d) *Interpretation.* Standing in $\bar{s}_1$, the four numbers are the prices of one unit of consumption delivered next period if the state is $\bar{s}_1 / \bar{s}_2$, and two periods ahead if the state is $\bar{s}_1 / \bar{s}_2$. Their row sums are the one- and two-period risk-free bond prices.

Inference: yes, for λ, δ, β (and κ , though not $\gamma, \bar{y}_1, \bar{y}_2$ separately). Let $a = Q_{11}^{(1)}$, $b = Q_{12}^{(1)}$, $c = Q_{11}^{(2)}$, $d = Q_{12}^{(2)}$. Then:

$$a = \beta\lambda, \quad \frac{d}{b} = \beta(\lambda + \delta) \Rightarrow e \equiv \frac{d}{b} - a = \beta\delta,$$

$$c = \beta^2\lambda^2 + (\beta - \beta\lambda)(\beta - \beta\delta) = a^2 + (\beta - a)(\beta - e).$$

So β solves $\beta^2 - (a + e)\beta + ae - (c - a^2) = 0$. Since $c - a^2 = \beta^2(1-\lambda)(1-\delta) > 0$, one root exceeds $\max(a, e)$ and the other lies below $\min(a, e)$. β must exceed both $a = \beta\lambda$ and $e = \beta\delta$, so it is the larger root:

$$\beta = \frac{(a + e) + \sqrt{(a - e)^2 + 4(c - a^2)}}{2}, \quad \lambda = \frac{a}{\beta}, \quad \delta = \frac{e}{\beta}, \quad \kappa = \frac{b}{\beta(1 - \lambda)}.$$

This is 8.24(b) (row recovery) followed by 8.24(d) (Perron root), done by hand. Contrast 8.26: risk-free bond prices alone are not enough.

Numerical exercise — see the companion code

$\lambda = .85, \delta = .7, \beta = .96, \gamma = 2, \bar{y} = (1, 1.2)$: observations (.816, .1, .707328, .1488). Then $e = .672$, roots .96 and .528, and the recovered values are $\beta = .96, \lambda = .85, \delta = .7, \kappa = .694444 = 1.2^{-2}$. The Perron root of the full $Q^{(1)}$ is also .96.

Companion code

Every numerical result above is reproduced by the notebook `ls_ch08.ipynb` in this folder (executed, outputs saved). Each claim is checked by an independent route with an `assert`.

Section	What it checks
Shared	Markov helpers: kernel $\beta P_{ij} \mu_j / \mu_i$, (8.10.3) by explicit double sum, history enumeration, stationary distribution. The ch. 5/7 LQ toolkit is not needed (no LQ problem here).
8.1–8.3	Closed forms vs SLSQP; $\partial W / \partial s = \theta$; decentralised consumers; v'_θ and v''_θ by finite differences.
8.4–8.7	Matrix prices vs enumeration of all histories; $(I - K)^{-1}$ vs truncated series; Q^h vs (8.10.3); Arrow budgets with $a = \Upsilon$.
8.8	Tax smoothing: closed form vs SLSQP over state-contingent revenue; government budget in both states.
8.9–8.10	Budgets and KKT conditions on 4000 periods; interest rates; sequential budgets.
8.11–8.13	$\tilde{\pi}$ sums to one; three representations of a_0 ; early-resolution holdings and debt limits.
8.14	Shares by inverse vs truncated sum; bond spanning; alternation closed form; default region (closed form vs utility sums; figure).
8.15–8.19	<code>brentq</code> on budgets; Euler equations of both consumers; binding natural debt limits.
8.20–8.23	Risk-free rates; entropy inequality; simulated beliefs selection (figure); recursive Pareto by binomial sums vs FOCs and envelope (figure).
8.24–8.28	Perron recovery; non-identification families (8.25, 8.26); corner equilibria in a Markov economy; quadratic-in- β recovery.

Figures are saved to `fig/ls08_*.pdf` with TrueType (not Type 3) fonts.